

Prompt Guide

How to write effective prompts for the Agent API.

The Agent API runs a bounded multi-turn loop: on each turn the model can call a tool (such as `web_search`), read the result, and decide whether to continue or answer. Prompts that work well with single-shot LLMs often underperform here, because the same text shapes tool selection, search query generation, and final response together.

Two parameters drive most of the prompt design:

- **instructions** sets the role, tone, formatting, and grounding rules that apply regardless of the user's question.
- **input** holds the actual question. It also seeds the first search query, so specificity here directly improves retrieval.

For hard constraints on retrieval (allowed domains, date ranges, region) and on the loop itself (max steps), use request parameters rather than prose. The sections below cover when to reach for each.

Instructions

Use the `instructions` parameter for role, tone, language, formatting, and grounding rules. Instructions apply on every turn of the agent loop, so put things here that hold regardless of the user's question.

Setting `instructions` with a preset **replaces** the preset's system prompt — it does not append. Each preset (`fast`, `low`, `medium`) already covers tool-call discipline, query construction, citation, and formatting, so the preset's prompt should be overridden only when app-specific behavior is needed. Without a preset, `instructions` is the only system prompt the model sees.

Example instructions block:

```
```text Instructions theme={null} You are a financial analyst writing for retail investors.
```

Rules: - Aim for brief sentences and paragraphs. - Define jargon the first time you use it. - Prefer concrete numbers over vague qualifiers ("up 12% YoY" not "growing strongly").

Grounding rules: - Cite sources inline by domain, e.g. (reuters.com). Do not write full URLs. - If searches return no relevant results after trying alternative phrasings, or if the only

matches are off-topic (different company, different fiscal year, etc.), say so explicitly rather than substituting related results.

Keep ``instructions`` focused. They are re-read on every turn of the agent loop, so bloat compounds across tool calls. If your block is growing long, check whether parts of it would be better expressed as request parameters: use `[`response_format`](/docs/agent-api/output-control)` with a JSON schema for machine-readable output, `[`web_search` filters](/docs/agent-api/tools/web-search#filters)` for retrieval constraints, or move query-specific framing into ``input``.

Built-in tools like ``web_search`` and ``fetch_url`` are tuned to work well without prompt-side guidance. You don't need to describe what they do, when to call them, or how to construct queries. Adjust tool-call count with the ``max_steps`` parameter and search constraints with ``web_search` filters`. If you're using custom ``instructions`` and want to nudge how the model uses built-in tools, you can reference them there as well.

For custom function tools you define yourself, the model relies on the ``description`` and parameter schema you provide, so make those as clear as you can. You can reinforce the tool's role in ``instructions`` if the description alone isn't enough to steer behavior.

## ## Input

Use the ``input`` parameter for the actual query you want answered. Input strongly shapes search behavior, so descriptive and specific phrasing directly improves retrieval. Vague inputs lead to vague searches.

**\*\*Example user prompt:\*\***

```
```text Input theme={null}
What are the best sushi restaurants in the world currently?
```

API Example

```
```python Python theme={null} from perplexity import Perplexity
```

```
client = Perplexity()
```

```
response = client.responses.create(preset="low", input="Explain how hosted LLM API pricing is typically structured: input vs output tokens, context window limits, and the implication for long-document workloads.", instructions="You are a concise, well-researched assistant. If searches still return no relevant results after trying alternative phrasings, say so explicitly rather than guessing.")
```

```
print(response.output_text) ``
```

```
``typescript Typescript theme={null} import Perplexity from '@perplexity-ai/perplexity_ai';
```

```
const client = new Perplexity();
```

```
const response = await client.responses.create({ preset: "low", input: "Explain how hosted LLM API pricing is typically structured: input vs output tokens, context window limits, and the implication for long-document workloads.", instructions: "You are a concise, well-researched assistant. If searches still return no relevant results after trying alternative phrasings, say so explicitly rather than guessing." });
```

```
console.log(response.output_text); ``
```

```
bash curl theme={null} curl https://api.perplexity.ai/v1/agent \ -H
"Authorization: Bearer $PERPLEXITY_API_KEY" \ -H "Content-Type:
application/json" \ -d '{ "preset": "low", "input": "Explain how
hosted LLM API pricing is typically structured: input vs output tokens, context
window limits, and the implication for long-document workloads.",
"instructions": "You are a concise, well-researched assistant. If searches still
return no relevant results after trying alternative phrasings, say so explicitly
rather than guessing." }' | jq
```

```
json theme={null} { "id": "resp_9ad22494-e7e4-4f84-a5b1-09a339d6bf79",
"created_at": 1779391834, "model": "openai/gpt-5.6-sol", "object":
"response", "output": [{ "results": [{
"id": 1, "snippet": "Every LLM API charges separately for **input
tokens** (your prompt, system message, and any context) and **output tokens**
(the model's response).\nOutput tokens always cost more – typically 2–5x the
input price.\n...\n**Key takeaway:** Context window size matters for
cost.\nModels with larger windows (Gemini's 1–2M tokens) let you send more
context per request, but more context means more input tokens billed.",
"title": "How LLM Token Pricing Works: A Complete Guide to API Costs in ...",
"url": "https://benchlm.ai/blog/posts/llm-token-pricing", "date":
"2026-03-26", "last_updated": "2026-05-20", "source":
"web" }, {
"id": 2, "snippet":
"Token usage is tracked in several categories:\n– **Input tokens** – tokens in
your request.\n– **Output tokens** – tokens generated in the response.\n...\nAPI
usage is priced per token, varying by model and whether tokens are input,
output, or cached.\nSee OpenAI's pricing page for current rates.",
"title": "What are tokens and how to count them? – OpenAI Help Center",
"url": "https://help.openai.com/en/articles/4936856-what-are-tokens-and-how-to-
count-them", "date": "2026-03-31", "last_updated":
"2026-03-31", "source": "web" }, {
```

```

{id": 3, "snippet": "When using extended thinking, all input and
output tokens, including the tokens used for thinking, count toward the context
window limit, with a few nuances in multi-turn situations.\n\nThe thinking budget
tokens are a subset of your `max_tokens` parameter, are billed as output tokens,
and count towards rate limits.\n...\n- **Token calculation:** All input and
output components count toward the context window, and all output components are
billed as output tokens.", "title": "Context windows - Claude API
Docs", "url": "https://platform.claude.com/docs/en/build-with-
claude/context-windows", "date": null, "last_updated":
"2026-05-21", "source": "web" }, {
{id": 4, "snippet": "Models like GPT-4.1 and Llama 4 Maverick
support context windows exceeding one million tokens, but larger windows do not
inherently raise per-token prices.", "title": "Understanding LLM
Cost Per Token: A 2026 Practical Guide", "url":
"https://www.silicondata.com/blog/llm-cost-per-token", "date":
"2026-05-07", "last_updated": "2026-05-18", "source":
"web" }, {
{id": 5, "snippet": "The
language there is \"**context window**\" = 128K and \"**max output tokens**\" =
16386", "title": "What is the maximum response length (output
tokens) for each GPT ...", "url":
"https://community.openai.com/t/what-is-the-maximum-response-length-output-
tokens-for-each-gpt-model/524066", "date": "2023-11-24",
"last_updated": "2026-05-19", "source": "web" }, {
{id": 6, "snippet": "Okay, so on pricing, now you're
going to be paying flat pricing irrespective how\n{ts:38} many tokens you are
using.\n\nSo, 900,000 tokens is going to be billed exactly the same\n{ts:44} as
9,000 tokens, which is a great deviation from the pricing tiers that Anthropic
Frontier Labs uses for long\n{ts:52} context.\n...\n\nBut, if you start using more
than 200,000 tokens, it starts to change\n{ts:104} because OpenAI is charging
almost two times for input tokens and 1.5 times for output tokens.\n{ts:112}
Same is the case with Gemini, but Anthropic is now charging a flat rate
for\n{ts:120} any token irrespective of how much context you're using.",
"title": "Anthropic Just Solved Long Context - YouTube", "url":
"https://www.youtube.com/watch?v=0w-8dYXDym8", "date": "2026-03-16",
"last_updated": "2026-05-21", "source": "web" }, {
{id": 7, "snippet": "But if your workload fills the
context window, the 128K model's per-request cost is 4x higher.\n\nPaying for
context capacity you do not use is free; paying for capacity you fill is
not.\n...\n\nEvery major LLM API charges different rates for input and output
tokens.\n...\n\nTypical ratios:\n- **OpenAI GPT-4o:** Output is 4x more expensive
than input ($10 vs $2.50 per million tokens)\n- **Anthropic Claude Sonnet:**
Output is 5x more expensive than input ($15 vs $3 per million tokens)\n-

```

**\*\*Google Gemini 1.5 Pro:\*\*** Output is 4x more expensive than input (\$5 vs \$1.25 per million tokens)\n...\nIf your application uses RAG, the retrieved context dominates your input token count.\nFive retrieved document chunks averaging 500 tokens each add 2,500 tokens to every request.\nAt \$2.50 per million input tokens, that context retrieval costs \$2.50 per 1,000 requests – or \$2,500 per million requests.\n...\nInput tokens are the tokens you send to the model (your prompt, system instructions, and any retrieved context).\nOutput tokens are the tokens the model generates in its response.\nOutput tokens cost 2–4x more than input tokens because generation requires more compute per token.",  
"title": "LLM Token Pricing Comparison 2026 – Cost Per Million Tokens",  
"url": "https://myengineeringpath.dev/tools/llm-pricing-comparison/",  
"date": "2026-03-20", "last\_updated": "2026-03-20",  
"source": "web" }, { "id": 8,  
"snippet": "\$5.00 / 1M tokens\nCached input:\n\$0.50 / 1M tokens\nOutput:\n\$30.00 / 1M tokens\n...\n\$2.50 / 1M tokens\nCached input:\n\$0.25 / 1M tokens\n...\n\$0.75 / 1M tokens\nCached input:\n\$0.075 / 1M tokens\nOutput:\n\$4.50 / 1M tokens", "title": "API Pricing – OpenAI",  
"url": "https://openai.com/api/pricing/",  
"date": "2026-04-09", "last\_updated": "2026-05-19",  
"source": "web" }, { "id": 9,  
"snippet": "The culprit is usually hiding in plain sight: output and reasoning tokens cost two to six times more than input tokens, and most teams don't realize how quickly they add up.\n...\nInput tokens include everything you send to the model: your prompt, system instructions, context, and conversation history.\nYou pay for every token the model \"reads,\" even if it doesn't use all of it.\n...\nOutput tokens typically cost two to four times more than input tokens.\n...\nOutput tokens require autoregressive generation, meaning the model runs once per token, sequentially.\n...\nThe pricing hierarchy follows a clear pattern: reasoning tokens are most expensive, followed by output tokens, with input tokens being least expensive.\n...\nContext windows grow over time as conversations continue or as you add more repository context.\nEach API call includes the full context, so longer contexts mean more input tokens per request.\nA code review that starts with 500 tokens of context might grow to 5,000 tokens as you add file history, related files, and conversation history.",  
"title": "Why Output & Reasoning Tokens Inflate LLM Costs (2026 Guide)",  
"url": "https://www.codeant.ai/blogs/input-vs-output-vs-reasoning-tokens-cost",  
"date": "2025-05-07", "last\_updated": "2026-05-15",  
"source": "web" }, { "id": 10,  
"snippet": "The rate limiter considers both input and output tokens.\n...\nThe AI model has a context window length, the maximum memory of tokens, an area for both processing your language input and for forming a response.\nStandard high-quality gpt-4 has a context length of 8k tokens.\ngpt-4-turbo has a context

length of 125k for understanding, with a limited output.\n...\nAs for rate limits: \*\*At tier 1\*\* (paying less than \$50 in the past),\n- `gpt-4-turbo-preview` has a limit of 150000 tokens per minute.\nHowever it has a much more restrictive 500000 tokens per day.\n- gpt-4 has a limit of 10000 tokens per minute; no daily limit.", "title": "Inputs tokens limit, data extraction - OpenAI Developer Community", "url": "https://community.openai.com/t/inputs-tokens-limit-data-extraction/612242", "date": "2024-02-03", "last\_updated": "2026-04-24", "source": "web" }, { "id": 11, "snippet": "Two concepts explain almost everything: the \*\*token\*\* (the chunks of text an AI reads and writes - roughly 3/4 of a word per token) and the \*\*context window\*\* (how much text the model can consider at once).\n...\n- \*\*You're paying for the whole window every time.\*\* Input cost is per-token; if your prompt includes 500k tokens of context, you pay for 500k input tokens on every call.\n...\nLLM API pricing has two prices, almost always: \*\*input\*\* (the tokens you send) and \*\*output\*\* (the tokens the model generates).\nOutput tokens cost more than input tokens - typically 3-5x more.\nWhy?\nOutput is where the model's expensive computation happens.\nEach output token requires the model to "think" about everything that came before it.\nInput tokens are largely a one-time cost - the model reads them, builds an internal representation, and then can produce many output tokens against that representation.", "title": "Tokens, context windows, and what they cost - Cyberax", "url": "https://cyberax.com/ai-playbook/tokens-context-windows-and-cost", "date": "2026-05-11", "last\_updated": "2026-05-17", "source": "web" }, { "id": 12, "snippet": "Its most notable feature is its 100k context window.\n...\n\*\*GPT-4\*\* (OpenAI) costs per 1,000 tokens:\*\*\n- Cost of Prompt: \$0.03\n- Cost of Completion: \$0.06\n\*\*Claude 2 (Anthropic) costs per 1,000 tokens:\*\*\n- Cost of Prompt: \$0.01102\n- Cost of Completion: \$0.03268", "title": "Using Anthropic: Best Practices, Parameters, and Large Context ...", "url": "https://www.prompthub.us/blog/using-anthropic-best-practices-parameters-and-large-context-windows", "date": "2025-01-15", "last\_updated": "2026-05-15", "source": "web" } ], "type": "search\_results", "queries": [ "LLM API pricing input vs output tokens context window", "OpenAI API pricing input output tokens context limit", "Anthropic pricing context window long documents" ] }, { "id": "msg\_30926530-7c19-4e04-98b9-75ebae7f593b", "content": [ { "text": "Hosted LLM APIs generally charge separately for \*\*input\*\* tokens (everything you send) and \*\*output\*\* tokens (everything the model generates), both subject to a hard context-window limit that caps how much text can be processed in a single call and strongly shapes the economics of long-document workloads.[web:1][web:2]" } ] }

[web:3][web:11] For long documents, the dominant cost is usually repeatedly paying for large input contexts rather than the model's answer itself, unless the responses are very long or "reasoning-heavy."

[web:7][web:9][web:11]\n\n## Input vs output tokens\n\nMost APIs meter usage as "tokens in" and "tokens out," with different prices for each.

[web:1][web:2][web:7][web:11] A token is a small chunk of text (often  $\sim \frac{3}{4}$  of an English word), so both prompts and responses are measured in hundreds to thousands of tokens rather than characters.

[web:11]\n\nCommon patterns:\n\n- Input tokens:\n - Include system prompt, user prompt, conversation history, and any retrieved context (e.g., RAG documents).

[web:2][web:7][web:9]\n - Are billed per million (or thousand) tokens at the lower rate tier.

[web:7][web:8][web:11]\n- Output tokens:\n - Include the final answer and any intermediate "thinking" or reasoning traces where supported.

[web:3][web:9]\n - Typically cost 2–5× more than input tokens because each generated token requires a full forward pass conditioned on all prior tokens.

[web:1][web:7][web:9][web:11]\n\nSome platforms also discount "cached" or reused input tokens that the model has already processed, which can matter when you reuse the same system prompt or fixed context across many calls.

[web:2][web:8]\n\n## Context window limits\n\nThe context window is the maximum total number of tokens (input plus output) the model can consider in a single request.

[web:3][web:10][web:11] Modern flagship models support windows from ~128K tokens up to 1M+ tokens, but the entire prompt plus the planned answer must fit within that limit.

[web:4][web:5][web:7][web:11]\n\nKey implications:\n\n- All tokens count: every input token (prompt, instructions, retrieved chunks, chat history) and every output token must fit inside the context window.

[web:3][web:10][web:11]\n- Per-request ceilings: models may expose both a context-window maximum and a separate "max output tokens" cap, so long inputs reduce the available budget for the answer.

[web:3][web:5]\n- Bigger windows do not automatically mean higher per-token prices, but they enable much larger prompts, so \*actual\* per-call cost scales with how full you make the window.

[web:1][web:4][web:7][web:11]\n\n## Pricing patterns across providers\n\nWhile numbers vary by vendor and model, the structure is very similar across major APIs.

[web:7][web:8][web:12]\n\nTypical patterns:\n\n- Dual pricing:\n - Separate \$/M input tokens and \$/M output tokens lines for each model.

[web:7][web:8][web:12]\n - Output rate commonly 2–5× the input rate.

[web:1][web:7][web:9][web:11]\n- Specialized token types:\n - Some providers expose "reasoning," "thinking," or "extended-thinking" tokens that are charged at a premium compared with normal output.

[web:3][web:9]\n - Cached or previously-seen input tokens may be billed at a discounted rate when the platform supports retrieval from a cache.

[web:2][web:8]\n- Long-context options:\n - Higher-end models offer larger context windows (hundreds of thousands to >1M tokens) at similar or slightly different per-token rates, but using the full window multiplies the cost of each call.

[web:4][web:7][web:11]\n\nThere are also emerging flat-pricing or "banded"

schemes for ultra-long-context tiers, but the common denominator remains that cost scales directly with tokens processed, regardless of how they are packaged.

[web:6][web:7]\n\n## Long-document workloads\n\nFor workloads like summarization, Q&A over corpora, or codebase analysis, the interaction between per-token pricing and context limits is crucial.[web:7][web:9][web:11]\n\nCore implications:\n\n- You pay repeatedly for context:\n - When doing retrieval-augmented generation (RAG), every query includes retrieved chunks, so the retrieved context often dominates the input token count.[web:7][web:9][web:11]\n - For example, five 500-token chunks (2,500 tokens) per request, at \$2.50 per million input tokens, costs ~\$2.50 per 1,000 calls just for those chunks, or ~\$2,500 per million calls.[web:7]\n - "Fill the window" gets expensive:\n - If you regularly send prompts close to a 128K-1M token context limit, each call can be tens to hundreds of times more expensive than a short-prompt call, even at the same per-token price.[web:4][web:7][web:11]\n - Long chat sessions or accreting code review threads can quietly grow context size over time, increasing cost per request as more history is resent each time.[web:9][web:11]\n - Trade-offs in chunking vs. single-shot:\n - Splitting a huge document into multiple smaller prompts can keep each call well under the context limit, at the cost of extra orchestration and possibly some loss of global coherence.[web:11]\n - Using a single enormous context window is simpler but can be much more expensive if every query resends most of the document.[web:1][web:7][web:11]\n\nFor practical system design, this means the main cost levers for long-document use cases are: how aggressively you limit retrieved context per request, how you manage chat or history growth, and how long you allow model outputs to be, given that all three are bounded by the context window and billed per token.[web:7][web:9][web:11]",

```
{
 "type": "output_text",
 "annotations": [],
 "logprobs": [],
 "role": "assistant",
 "status": "completed",
 "type": "message"
}, {
 "status": "completed",
 "error": null,
 "usage": {
 "input_tokens": 6297,
 "output_tokens": 1408,
 "total_tokens": 7705,
 "cost": {
 "currency": "USD",
 "input_cost": 0.01085,
 "output_cost": 0.02816,
 "total_cost": 0.04294,
 "cache_creation_cost": null,
 "cache_read_cost": 0.00143,
 "tool_calls_cost": 0.0025
 },
 "input_tokens_details": {
 "cache_creation_input_tokens": 0,
 "cache_read_input_tokens": 3584,
 "cached_tokens": 3584
 },
 "tool_calls_details": {
 "search_web": {
 "invocation": 1
 }
 },
 "output_tokens_details": {
 "reasoning_tokens": 0
 }
 },
 "background": false,
 "completed_at": 1779391834,
 "frequency_penalty": 0,
 "incomplete_details": null,
 "instructions": "## Abstract\n<role>\n\nYou are an AI assistant developed by Perplexity AI. Given a user's query, your goal is to generate an expert, useful, factually correct, and contextually relevant"
```

response by leveraging available tools and conversation history. First, you will receive the tools you can call iteratively to gather the necessary knowledge for your response. You need to use these tools rather than using internal knowledge. Second, you will receive guidelines to format your response for clear and effective presentation. Third, you will receive guidelines for citation practices to maintain factual accuracy and credibility.

**Instructions**

**<tools\_workflow>** Begin each turn with tool calls to gather information. You must call at least one tool before answering, even if information exists in your knowledge base. Decompose complex user queries into discrete tool calls for accuracy and parallelization. After each tool call, assess if your output fully addresses the query and its subcomponents. Continue until the user query is resolved or until the **<tool\_call\_limit>** below is reached. End your turn with a comprehensive response. Never mention tool calls in your final response as it would badly impact user experience.

**<tool\_call\_limit>** Make at most three tool calls before concluding.

**Citation Instructions**

Your response must include at least 1 citation. Add a citation to every sentence that includes information derived from tool outputs. Tool results are provided using ``id`` in the format ``type:index``. ``type`` is the data source or context. ``index`` is the unique identifier per citation.

**<common\_source\_types>** are included below.

**<common\_source\_types>**

- ``web``: Internet sources
- ``page``: Full web page content
- ``conversation_history``: past queries and answers from your interaction with the user

**<formatting\_citations>**

Use brackets to indicate citations like this: `[type:index]`. Commas, dashes, or alternate formats are not valid citation formats. If citing multiple sources, write each citation in a separate bracket like `[web:1][web:2][web:3]`.

Correct: `"The Eiffel Tower is in Paris [web:3]."`

Incorrect: `"The Eiffel Tower is in Paris [web-3]."`

**<formatting\_citations>**

Your citations must be inline – not in a separate References or Citations section. Cite the source immediately after each sentence containing referenced information. If your response presents a markdown table with referenced information from ``web``, ``memory``, ``attached_file``, or ``calendar_event`` tool result, cite appropriately within table cells directly after relevant data instead in of a new column. Do not cite ``generated_image`` or ``generated_video`` inside table cells.

**Response Guidelines**

**<response\_guidelines>**

Responses are displayed on web interfaces where users should not need to scroll extensively. Limit responses to 5 sections maximum. Users can ask follow-up questions if they need additional detail. Prioritize the most relevant information for the initial query.

**Answer Formatting**

- Begin with a direct 1–2 sentence answer to the core question.
- Organize the rest of your answer into sections led with Markdown headers (using `##`, `###`) when appropriate to ensure clarity (e.g. entity

definitions, biographies, and wikis).\n- Your answer should be at least 3 sentences long.\n- Each Markdown header should be concise (less than 6 words) and meaningful.\n- Markdown headers should be plain text, not numbered.\n- Between each Markdown header is a section consisting of 2-3 well-cited sentences.\n- When comparing entities with multiple dimensions, use a markdown table to show differences (instead of lists).\n- Whenever possible, present information as bullet point lists to improve readability.\n- You are allowed to bold at most one word (**example**) per paragraph. You can't bold consecutive words.\n- For grouping multiple related items, present the information with a mix of paragraphs and bullet point lists. Do not nest lists within other lists.\n\n### Tone\n<tone>\nExplain clearly using plain language. Use active voice and vary sentence structure to sound natural. Ensure smooth transitions between sentences. Avoid personal pronouns like "I". Keep explanations direct; use examples or metaphors only when they meaningfully clarify complex concepts that would otherwise be unclear.\n</tone>\n\n### Lists and Paragraphs\n<lists\_and\_paragraphs>\nUse lists for: multiple facts/recommendations, steps, features/benefits, comparisons, or biographical information.\n\nAvoid repeating content in both intro paragraphs and list items. Keep intros minimal. Either start directly with a header and list, or provide 1 sentence of context only.\n\nList formatting:\n- Use numbers when sequence matters; otherwise bullets (-) with a space after the dash.\n- Use numbers when sequence matters; otherwise bullets (-).\n- No whitespace before bullets (i.e. no indenting), one item per line.\n- Sentence capitalization; periods only for complete sentences.\n\nParagraphs:\n- Use for brief context (2-3 sentences max) or simple answers\n- Separate with blank lines\n- If exceeding 3 consecutive sentences, consider restructuring as a list\n</lists\_and\_paragraphs>\n\n### Summaries and Conclusions\n<summaries\_and\_conclusions>\nAvoid summaries and conclusions. They are not needed and are repetitive. Markdown tables are not for summaries. For comparisons, provide a table to compare, but avoid labeling it as 'Comparison/Key Table', provide a more meaningful title.\n</summaries\_and\_conclusions>\n\n## Prohibited Meta-Commentary\n<prohibited\_commentary>\n- Never reference your information gathering process in your final answer.\n- Do not use phrases such as:\n- "Based on my search results..." \n- "Now I have gathered comprehensive information..." \n- "According to my research..." \n- "My search revealed..." \n- "I found information about..." \n- "Let me provide a detailed answer..." \n- "Let me compile this information..." \n- "Short Answer: ..."\n- Begin answers immediately with factual content that directly addresses the user's query.\n</prohibited\_commentary>\n\n<copyright\_requirements>\n- Never reproduce copyrighted content (text, lyrics, etc.)\n- You may share public domain content (expired copyrights, traditional works)\n- When copyright status is uncertain, treat as copyrighted\n- Keep summaries brief (under 30 words) and

```
original – don't reconstruct sources\n– Brief factual statements (names, dates, facts) are always acceptable\n</copyright_requirements>\n\nCurrent date: Thursday, May 21, 2026\n\n", "max_output_tokens": 8192, "max_tool_calls": null, "metadata": {}, "parallel_tool_calls": true, "presence_penalty": 0, "previous_response_id": null, "prompt_cache_key": null, "reasoning": null, "safety_identifier": null, "service_tier": "default", "store": true, "temperature": 1, "text": { "format": { "type": "text" }, "tool_choice": "auto", "tools": [{ "type": "web_search" }, { "type": "fetch_url" }], "top_logprobs": 0, "top_p": 1, "truncation": "disabled", "user": null }
```

## Best Practices

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Use natural language, but include the vocabulary and context that would actually appear on relevant pages. Add a few words of context to disambiguate when a term could mean multiple things. Specificity in `input` directly improves retrieval.

**\*\*Good Example\*\*:** "Compare energy efficiency ratings of heat pumps vs. traditional HVAC for residential use"

**\*\*Poor Example\*\*:** "Tell me which home heating is better"

If you want a list, say how long. Without an explicit cap, the model picks an arbitrary length.

**\*\*Good Example\*\*:** "List the top 5 sushi restaurants in Tokyo"

**\*\*Poor Example\*\*:** "Give me a list of sushi restaurants"

Can be useful if you want to nudge how the model handles tool output. Things like citation style, grounding behavior, or response formatting fit naturally here, since instructions apply on every turn of the agent loop.

**\*\*Example\*\*** (``instructions``): "Cite sources inline by domain (e.g., reuters.com). State explicitly when tool results don't fully answer the question."

## Reading Sources from the Response

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Read URLs and source metadata from the response payload, not from the model's written answer. For non-streaming responses, search results are inside `response.output[]` as items where `type == "search_results"` — the response contains one such item per search the model ran, all sharing a single citation `id` space, so collect the results from every item rather than only the first (multi-step presets like deep research run many searches). Pull URLs from `results[].url`. For streaming, listen for `response.reasoning.search_results` events (again, one per search). See [Output Control](#) for the full response shape.

The model has access to URLs from tool output and can include them in its response if asked, but it's prone to mistyping or paraphrasing them. Presets also configure a citation style for the model — an index that maps to the search result `id` (e.g., `[1]` or `[web:1]`) or inline links, depending on the preset — so asking for URLs in prose fights the default citation format. Treat the model's text as the prose answer and the structured `search_results` items as the authoritative source list.

## Reduce Hallucinations

---

LLMs are tuned to be helpful, which can occasionally lead them to provide an answer when search results are thin or off-target rather than flagging the gap. The agent loop helps, since the model can refine queries and search again, but it does not eliminate the failure modes. Hallucination is most likely when the information isn't web-accessible (posts that require authentication, private documents, paywalled content), when repeated searches return related but non-matching results, or when very recent information isn't indexed yet.

A few short additions to `instructions` cover most of these cases. Grounding rules belong here because instructions are re-read on every turn of the agent loop, so the same rule applies to the first search and to any follow-ups.

**Give the model permission to say it didn't find anything.** With an explicit out, the model is more likely to acknowledge insufficient results instead of leaning on training data to fill the gap.

```
text Instructions theme={null} If searches do not return relevant results after trying alternative phrasings, say so explicitly rather than providing speculative information.
```

**Require disclosure of near-misses.** When search returns related but non-matching results (a different year, a parent company instead of a subsidiary, a similar product), asking the model to surface the mismatch up front keeps these cases from being presented as direct answers.

text Instructions theme={null} If you find related but non-matching results (for example, a different year, a parent company, or a subsidiary), state the mismatch explicitly before answering.

## Use Parameters, Not Prose, for Hard Constraints

For source, date, or region constraints, prefer the `web_search` parameters over describing the constraint in prose. Parameters are applied by the search backend on every call, while prose-based filters are interpreted by the model and may not carry through every turn of the loop.

Keep `input` focused on the question itself, and move structural constraints into the tool config:

```
python Avoid theme={null} client.responses.create(preset="low",
input="Using only Wikipedia as a source, summarize the history of the Apollo
program: each crewed mission, their objectives, and key outcomes.")
```

```
json theme={null} { "id": "resp_50148504-902d-4f26-8432-2930506d13af",
"created_at": 1779895991, "model": "openai/gpt-5.6-sol", "object":
"response", "output": [{ "results": [{
"id": 1, "snippet": "The Apollo program, also known as Project
Apollo, was the United States human spaceflight program led by NASA, which
landed the first humans on the Moon in 1969.\nApollo was conceived in 1960 in
the Dwight D. Eisenhower presidency during Project Mercury and executed after
Project Gemini.\nApollo was later dedicated to President John F. Kennedy's
national goal, \"before this decade is out, of landing a man on the Moon and
returning him safely to the Earth\" in his address to the U.S. Congress on May
25, 1961.\n...\nApollo ran from 1961 to 1972, with the first crewed flight in
1968.", "title": "Apollo program - Wikipedia", "url":
"https://en.wikipedia.org/wiki/Apollo_program", "date": "2001-09-
24", "last_updated": "2026-05-21", "source": "web"
}, { "id": 2, "snippet": "Apollo 7
(October 11-22, 1968) was the first crewed flight in NASA's Apollo program, and
saw the resumption of human spaceflight by the agency after the fire that killed
the three Apollo 1 astronauts during a launch rehearsal test on January 27,
1967.", "title": "Apollo 7 - Wikipedia", "url":
"https://en.wikipedia.org/wiki/Apollo_7", "date": "2001-08-27",
"last_updated": "2026-05-27", "source": "web"
},
{ "id": 3, "snippet": "Apollo 8 (December 21-27,
1968) was the first crewed spacecraft to leave Earth's gravitational sphere of
influence, and the first human spaceflight to reach the Moon.\nThe crew orbited
```

the Moon ten times without landing and then returned to Earth.\n

The three astronauts—Frank Borman, Jim Lovell, and William Anders—were the first humans to see and photograph the far side of the Moon and an Earthrise.\n

Apollo 8 launched on December 21, 1968, and was the second crewed spaceflight mission flown in the United States Apollo space program (the first, Apollo 7, stayed in Earth orbit).\n

Apollo 8 was the third flight and the first crewed launch of the Saturn V rocket.\n

It was the first human spaceflight from the Kennedy Space Center, adjacent to Cape Kennedy Air Force Station in Florida.", "title": "Apollo 8 – Wikipedia", "url": "https://en.wikipedia.org/wiki/Apollo\_8", "date": "2001-03-17", "last\_updated": "2026-05-22", "source": "web", }, { "id": 4, "snippet": "\*\*Apollo 9\*\* (March 3–13, 1969) was the third human spaceflight in NASA's Apollo program, which successfully tested systems and procedures critical to landing on the Moon.\n

The three-man crew consisted of Commander James McDivitt, Command Module Pilot David Scott, and Lunar Module Pilot Rusty Schweickart.\n

Flown in low Earth orbit, it was the second crewed Apollo mission that the United States launched via a Saturn V rocket, and was the first flight of the full Apollo spacecraft: the command and service module (CSM) with the Lunar Module (LM).", "title": "Apollo 9 – Wikipedia", "url": "https://en.wikipedia.org/wiki/Apollo\_9", "date": "2001-08-27", "last\_updated": "2026-05-22", "source": "web", }, { "id": 5, "snippet": "\*\*Apollo 10\*\* (May 18–26, 1969) was the fourth human spaceflight in the United States' Apollo program and the second to orbit the Moon.\n

NASA, the mission's operator, described it as a \"dress rehearsal\" for the first Moon landing (Apollo 11, two months later).", "title": "Apollo 10 – Wikipedia", "url": "https://en.wikipedia.org/wiki/Apollo\_10", "date": "2001-09-22", "last\_updated": "2026-05-22", "source": "web", }, { "id": 6, "snippet": "The Apollo program was a United States human spaceflight program carried out from 1961 to 1972 by the National Aeronautics and Space Administration (NASA), which landed the first astronauts on the Moon.\n

The program used the Saturn IB and Saturn V launch vehicles to lift the Command/Service Module (CSM) and Lunar Module (LM) spacecraft into space, and the Little Joe II rocket to test a launch escape system which was expected to carry the astronauts to safety in the event of a Saturn failure.", "title": "List of Apollo missions – Wikipedia", "url": "https://en.wikipedia.org/wiki/List\_of\_Apollo\_missions", "date": "2007-02-13", "last\_updated": "2026-05-10", "source": "web", }, { "id": 7, "snippet": "\*\*Many are familiar with Apollo 11, the mission that landed humans on the Moon for the first time.\n

It was part of the larger Apollo program.\n

\*\*\n

\*\*There were

several missions during the Apollo program from 1961 to 1972.\nHumans landed on the moon during six missions, Apollo 11, 12, 14, 15, 16, and 17.\n\*\*",

"title": "The Apollo Missions | National Air and Space Museum",

"url": "https://airandspace.si.edu/explore/stories/apollo-missions",

"date": "2021-11-04", "last\_updated": "2026-05-17",

"source": "web" }, { "id": 8,

"snippet": "\*\*Apollo 7\*\* was a mission in the NASA's Apollo program.\nIt was the first crewed mission in the Apollo program and the first crewed US space flight after Apollo 1 disaster.\nThe mission was a C type mission.\nApollo 7 was launched on October 11, 1968 and stayed in space for 10 days, 20 hours, 9 minutes and three seconds.\n...\nApollo 7 was the first crewed launch of the Saturn IB launch vehicle and the first three-person US space mission.\n...\nThe mission was designed to test the re-made Block II Apollo Command/Service Module.\n...\nThe mission was a success.\nIt gave NASA the confidence to launch Apollo 8 later.",

"title": "Apollo 7 - Simple English Wikipedia, the free encyclopedia", "url":

"https://simple.wikipedia.org/wiki/Apollo\_7", "date": "2012-02-29",

"last\_updated": "2026-05-26", "source": "web" },

{ "id": 9, "snippet": "\*\*Apollo 8\*\* was a mission in the Apollo program in December 1968.\nIt was the first crewed spaceflight to leave Earth orbit and first to orbit the Moon.\nCommander Frank Borman, Pilot Jim Lovell and Bill Anders transmitted a television show while they were in orbit.",

"title": "Apollo 8 - Simple English Wikipedia, the free encyclopedia",

"url": "https://simple.wikipedia.org/wiki/Apollo\_8", "date": "2012-01-20",

"last\_updated": "2026-05-23", "source": "web"

}, { "id": 10, "snippet": "\*\*Apollo 9\*\* was a mission in NASA's Apollo program.\nIt was the third crewed mission in the Apollo program and was the first flight of the Command/Service Module (CSM) with the Lunar Module (LM).\nThe crew was Commander James A. McDivitt, Command Module Pilot David R. Scott, and Lunar Module Pilot Russell L. Schweickart.",

"title": "Apollo 9 - Simple English Wikipedia, the free encyclopedia",

"url": "https://simple.wikipedia.org/wiki/Apollo\_9", "date": "2012-05-17",

"last\_updated": "2026-04-01", "source": "web"

}, { "id": 11, "snippet": "The national effort that enabled Astronaut Neil Armstrong to speak those words as he stepped onto the lunar surface fulfilled a dream as old as humanity.\nProject Apollo's goals went beyond landing Americans on the moon and returning them safely to Earth.\nThey included:\n- Establishing the technology to meet other national interests in space.\n- Achieving preeminence in space for the United States.\n- Carrying out a program of scientific exploration of the Moon.\n- Developing human capability to work in the lunar environment.\n...\nThe flight mode, lunar orbit rendezvous, was selected in 1962.\nThe boosters for the program were the

Saturn IB for Earth orbit flights and the Saturn V for lunar flights.\nApollo was a three-part spacecraft: the command module (CM), the crew's quarters and flight control section; the service module (SM) for the propulsion and spacecraft support systems (when together, the two modules are called CSM); and the lunar module (LM), to take two of the crew to the lunar surface, support them on the Moon, and return them to the CSM in lunar orbit.",

"title": "The Apollo Program – NASA", "url": "https://www.nasa.gov/the-apollo-program/", "date": "2023-03-23", "last\_updated": "2026-05-18", "source": "web" }, { "id": 12, "snippet": "\*\*On October 11, 1968 Apollo 7 was launched on a Saturn IB rocket, making it the first successful crewed Apollo mission and the only crewed Apollo mission to use the Saturn IB Rocket.\*\*\nApollo 7 was the first test of the command and service module with a crew.\nThe crew orbited the Earth 163 times and spent 10 days and 20 hours in space.\nThis mission was the first opportunity to test the first of the new Block II spacecraft (CSM 101) in orbit.\nThe only significant difficulty in the mission was the fact that all three astronauts developed severe head colds.",

"title": "Apollo 7 | National Air and Space Museum", "url": "https://airandspace.si.edu/explore/stories/apollo-missions/apollo-7", "date": "2021-07-29", "last\_updated": "2026-03-25", "source": "web" }, "type": "search\_results", "queries": [ "Apollo program Wikipedia", "List of Apollo missions Wikipedia", "Apollo 7 Wikipedia", "Apollo 8 Wikipedia", "Apollo 9 Wikipedia", "Apollo 10 Wikipedia" ] }, { "contents": [ { "snippet": "The Apollo program was a United States human spaceflight program carried out from 1961 to 1972 by the National Aeronautics and Space Administration (NASA), which landed the first astronauts on the Moon. The program used the Saturn IB and Saturn V launch vehicles to lift the Command/Service Module (CSM) and Lunar Module (LM) spacecraft into space, and the Little Joe II rocket to test a launch escape system which was expected to carry the astronauts to safety in the event of a Saturn failure. Uncrewed test flights beginning in 1966 demonstrated the safety of the launch vehicles and spacecraft to carry astronauts, and four crewed flights beginning in October 1968 demonstrated the ability of the spacecraft to carry out a lunar landing mission.\n\nApollo achieved the first crewed lunar landing on the Apollo 11 mission, when Neil Armstrong and Buzz Aldrin landed their LM \*Eagle\* in the Sea of Tranquility and walked on the lunar surface, while Michael Collins remained in lunar orbit in the CSM \*Columbia\*, and all three landed safely on Earth on July 24, 1969. Five subsequent missions landed astronauts on various lunar sites, ending in December 1972 with 12 men having walked on the Moon and 842 pounds (382 kg) of lunar rocks and soil samples returned to Earth, greatly contributing to the understanding of the

Moon's composition and geological history.

Two Apollo missions were failures: a 1967 cabin fire killed the entire Apollo 1 crew during a ground test in preparation for what was to be the first crewed flight; and the third landing attempt on Apollo 13 was aborted by an oxygen tank explosion en route to the Moon, which disabled the CSM \*Odyssey\*'s electrical power and life support systems, and made the propulsion system unsafe to use. The crew circled the Moon and were returned safely to Earth using the LM \*Aquarius\* as a "lifeboat" for these functions.

### Uncrewed test flights

From 1961 through 1967, Saturn launch vehicles and Apollo spacecraft components were tested in uncrewed flights.

#### Saturn I

The Saturn I launch vehicle was originally planned to carry crewed Command Module flights into low Earth orbit, but its 20,000-pound (9,100 kg) payload capacity limit could not lift even a partially fueled Service Module, which would have required building a lightweight retrorocket module for deorbit. These plans were eventually scrapped in favor of using the uprated Saturn IB to launch the Command Module with a half-fueled Service Module for crewed Earth orbit tests. This limited Saturn I flights to Saturn launch vehicle development, CSM boilerplate testing, and three Pegasus micrometeoroid satellite missions in support of Apollo.

#### Saturn I missions

Mission	LV	Launch	Pad	Remarks	Refs
SA-1	SA-1	October 27, 1961, 15:06	LC-34	Test of Saturn I first stage S-I; dummy upper stages carried water	
SA-2	SA-2	April 25, 1962, 14:00	LC-34	Dummy upper stages released 22,900 U.S. gallons (86,685 L) of water into upper atmosphere, to investigate effects on radio transmission and changes in local weather conditions	
SA-3	SA-3	November 16, 1962, 17:45	LC-34	Repeat of SA-2 mission	
SA-4	SA-4	March 28, 1963, 20:11	LC-34	Test premature shutdown of a single S-I engine	
SA-5	SA-5	January 29, 1964, 16:25	LC-37B	First flight of live second stage. First orbital flight.	
AS-101	SA-6	May 28, 1964, 17:07	LC-37B	Tested first boilerplate Apollo command and service module (CSM) for structural integrity	
AS-102	SA-7	September 18, 1964, 17:22	LC-37B	Carried first programmable-in-flight computer on the Saturn I vehicle; last launch vehicle development flight	
AS-103	SA-9	February 16, 1965, 14:37	LC-37B	Carried Pegasus A satellite and boilerplate CSM	
AS-104	SA-8	May 25, 1965, 07:35	LC-37B	Carried Pegasus B satellite and boilerplate CSM	
AS-105	SA-10	July 30, 1965, 13:00	LC-37B	Carried Pegasus C satellite and boilerplate CSM	

There was some incongruity in the numbering and naming of the first three uncrewed Apollo-Saturn (AS) or Apollo flights. This is due to AS-204 being renamed to Apollo 1 posthumously. This crewed flight was to have followed the first three uncrewed flights. After the fire which killed the AS-204 crew on the pad during a test and training exercise, uncrewed Apollo flights resumed to test the Saturn V launch vehicle and the Lunar Module; these were designated Apollo 4, 5 and 6. The first crewed Apollo mission was thus Apollo 7. Simple "Apollo" numbers were never assigned to the first three uncrewed

flights, although renaming AS-201, AS-202, and AS-203 as Apollo 1-A, Apollo 2 and Apollo 3, had been briefly considered.

### Saturn IB

The Saturn I was converted to the Up-rated Saturn I, eventually designated Saturn IB, by replacing the S-IV second stage with the S-IVB, which would also be used as the third stage of the Saturn V with the addition of on-orbit restart capability. This increased the payload capacity to 46,000 pounds (21,000 kg), enough to orbit a Command Module with a half-fueled Service Module, and more than enough to orbit a fully fueled Lunar Module.

Two suborbital tests of the Apollo Block I Command and Service Module, one S-IVB development test, and one Lunar Module test were conducted. Success of the LM test led to cancellation of a planned second uncrewed flight.

### Saturn IB missions

Mission	LV	Launch	Pad	Remarks	Refs
AS-201	SA-201	February 26, 1966, 16:12	LC-34	First test of Saturn IB and Block I Apollo CSM. Suborbital flight landed the CM in the Atlantic Ocean, demonstrating the heat shield. Propellant pressure loss caused premature SM engine shutdown.	
AS-203	SA-203	July 5, 1966, 14:53	LC-37B	No Apollo spacecraft; instrumentation and video observed on-orbit behavior of S-IVB liquid hydrogen fuel in support of restart capability design for Saturn V. Deemed a success, despite inadvertent destruction of S-IVB during final overpressure tank rupture test.	
AS-202	SA-202	August 25, 1966, 17:15	LC-34	Suborbital flight to Pacific Ocean splashdown. CM heat shield tested to higher speed; successful SM firings.	
Apollo 5	SA-204	January 22, 1968, 22:48	LC-37B	First flight of LM successfully fired descent engine and ascent engines; demonstrated "fire-in-the-hole" landing abort test.	

### Launch escape system tests

From August 1963 to January 1966, a number of tests were conducted at the White Sands Missile Range for development of the launch escape system (LES). These included simulated "pad aborts", which might occur while the Apollo-Saturn space vehicle was still on the launch pad, and flights on the Little Joe II rocket to simulate Mode I aborts which might occur while the vehicle was in the air.

### Launch escape system tests

Mission	LV	Launch	Pad	Remarks	Refs
QTV	Little Joe II	August 28, 1963, 13:05	LC-36	Little Joe II qualification test	
Pad Abort Test 1	N/a	November 7, 1963, 16:00	LC-36	Launch escape system (LES) abort test from launch pad	
A-001	Little Joe II	May 13, 1964, 13:00	LC-36	LES transonic test, success except for parachute failure	
A-002	Little Joe II	December 8, 1964, 15:00	LC-36	LES maximum altitude, Max-Q abort test	
A-003	Little Joe II	May 19, 1965, 13:01	LC-36	LES canard maximum altitude abort test	
Pad Abort Test 2	N/a	June 29, 1965, 13:00	LC-36	LES pad abort test of near Block-I CM	
A-004	Little Joe II	January 20, 1966, 15:17	LC-36	LES test of maximum weight, tumbling Block-I CM	

### Saturn V

Prior to George Mueller's tenure as NASA's Associate Administrator for Manned Space Flight starting in 1963, it was assumed that 20 Saturn Vs, with at least 10 unpiloted test flights, would be required to achieve

a crewed Moon landing, using the conservative one-stage-at-a-time testing philosophy used for the Saturn I. But Mueller introduced the "all-up" testing philosophy of using three live stages plus the Apollo spacecraft on every test flight. This achieved development of the Saturn V with far fewer uncrewed tests, enabling a Moon landing by the 1969 goal. The size of the Saturn V production lot was reduced from 20 to 15 units.

Three uncrewed test flights were planned to human-rate the super heavy-lift Saturn V which would take crewed Apollo flights to the Moon. Success of the first flight and qualified success of the second led to the decision to cancel the third uncrewed test.

**Saturn V missions**

Mission	LV	Launch	Pad	Remarks	Refs
Apollo 4	SA-501	November 9, 1967, 12:00	LC-39A	First flight of Saturn V rocket; successfully demonstrated S-IVB third stage restart and tested CM heat shield at lunar re-entry speeds.	
Apollo 6	SA-502	April 4, 1968, 16:12	LC-39A	Second flight of Saturn V; severe "pogo" vibrations caused two second-stage engines to shut down prematurely, and third stage restart to fail. SM engine used to achieve high-speed re-entry, though less than Apollo 4. NASA identified vibration fixes and declared Saturn V man-rated.	

**Alphabetical mission types**

The Apollo program required sequential testing of several major mission elements in the runup to a crewed lunar landing. An alphabetical list of major mission types was proposed by Owen Maynard in September 1967. Two "A-type" missions performed uncrewed tests of the CSM and the Saturn V, and one B-type mission performed an uncrewed test of the LM. The C-type mission, the first crewed flight of the CSM in Earth orbit, was performed by Apollo 7.

The list was revised upon George Low's proposal to commit a mission to lunar orbit ahead of schedule, an idea influenced by the status of the CSM as a proven craft and production delays of the LM. Apollo 8 was reclassified from its original assignment as a D-type mission, a test of the complete CSM/LM spacecraft in Earth orbit, to a "C-prime" mission which would fly humans to the Moon. Once complete, it eliminated the need for the E-type objective of a medium Earth orbital test. The D-type mission was instead performed by Apollo 9; the F-type mission, Apollo 10, flew the CSM/LM spacecraft to the Moon for final testing, without landing. The G-type mission, Apollo 11, performed the first lunar landing, the central goal of the program.

The initial A-G list was expanded to include later mission types: H-type missions—Apollo 12, 13 (planned) and 14—would perform precision landings on the lunar surface, and J-type missions—Apollo 15, 16 and 17—would perform thorough scientific investigation of the Moon from the lunar surface. The I-type mission, which called for extended scientific investigation of the Moon from lunar orbit, was incorporated into the J-type missions.

**Alphabetical mission types of the Apollo Program**

Type	Mission	Description
A	Apollo 4 – Apollo 6	Uncrewed flights of launch vehicles and the CSM, to demonstrate its design and to certify its safety for humans.
B	Apollo 5	Uncrewed flight of the LM to

demonstrate its design and to certify its safety for humans.

C|Apollo 7|Crewed flight demonstration of CSM in low Earth orbit.

C'|Apollo 8|Crewed flight demonstration of CSM in lunar orbit.

D|Apollo 9|Crewed flight demonstration of CSM and LM in low Earth orbit, operating the equipment together in space and (insofar as possible in Earth orbit) performing the maneuvers involved in a lunar landing.

E|N/a|Crewed flight demonstration of CSM and LM in medium Earth orbit, performing the maneuvers involved in a lunar landing.

F|Apollo 10|Crewed flight demonstration of CSM and LM in lunar orbit, performing all G-type mission goals except for the final descent to and landing on the lunar surface.

G|Apollo 11|Crewed lunar landing demonstration.

H|- Apollo 12 - Apollo 13 (planned) - Apollo 14|Precision crewed lunar landing demonstration and systematic lunar exploration.

I|N/a|Extended scientific investigation of the Moon from lunar orbit. (Not used, incorporated into J type)

J|- Apollo 15 - Apollo 16 - Apollo 17|Extended scientific investigation of the Moon on the lunar surface and from lunar orbit.

### Crewed missions

The Block I CSM spacecraft did not have capability to fly with the LM, and the three crew positions were designated Command Pilot, Senior Pilot, and Pilot, based on U.S. Air Force pilot ratings. The Block II spacecraft was designed to fly with the Lunar Module, so the corresponding crew positions were designated Commander, Command Module Pilot, and Lunar Module Pilot regardless of whether a Lunar Module was present or not on any mission.

Seven of the missions involved extravehicular activity (EVA), spacewalks or moonwalks outside of the spacecraft. These were of three types: testing the lunar EVA suit in Earth orbit (Apollo 9), exploring the lunar surface, and retrieving film canisters from the Scientific Instrument Module stored in the Service Module.

\*\*Crewed missions\*\*

Mission	Patch	Launch date	Crew	Launch vehicle	CM name	LM name	Duration	Remarks	Refs
Apollo 1		February 21, 1967	Launch Complex 34 (planned)	Gus Grissom	Ed White	Roger B. Chaffee	Saturn IB (SA-204)	N/a	N/a
Never launched. On January 27, 1967, a fire in the command module during a launch pad test killed the crew and destroyed the module. This flight was originally designated AS-204, and was renamed to Apollo 1 at the request of the crew's families.									
Apollo 7		October 11, 1968, 15:02 GMT	Launch Complex 34	Wally Schirra	Donn F. Eisele	Walter Cunningham	Saturn IB (AS-205)	N/a	N/a
10 d 20 h 09 m 03 s Test flight of Block II CSM in Earth orbit; included first live TV broadcast from American spacecraft.									
Apollo 8		December 21, 1968, 12:51 GMT	Launch Complex 39A	Frank Borman	James Lovell	William Anders	Saturn V (SA-503)	N/a	N/a
06 d 03 h 00 m 42 s First humans to leave Earth orbit and first to arrive at the Moon, first circumlunar flight of CSM, had ten lunar orbits in 20 hours. First crewed flight of Saturn V.									
Apollo 9		March 3, 1969, 16:00 GMT	Launch Complex 39A	James McDivitt	David Scott	Rusty Schweickart	Saturn V (SA-504)	*Gumdrop*	*Spider*
10 d 01 h 00 m 54 s First crewed flight test of Lunar									

Module; tested propulsion, rendezvous and docking in Earth orbit. EVA tested the Portable Life Support System (PLSS).| |\n|Apollo 10| |May 18, 1969, 16:49 GMT Launch Complex 39B|Thomas P. Stafford John Young Eugene Cernan|Saturn V (SA-505)|\*Charlie Brown\*|\*Snoopy\*|08 d 00 h 03 m 23 s|\"Dress rehearsal\" for lunar landing. The LM descended to 8.4 nautical miles (15.6 km) from lunar surface.| |\n|Apollo 11| |July 16, 1969, 13:32 GMT Launch Complex 39A|Neil Armstrong Michael Collins Edwin \"Buzz\" Aldrin|Saturn V (SA-506)|\*Columbia\*|\*Eagle\*|08 d 03 h 18 m 35 s|First crewed landing in Sea of Tranquility (Tranquility Base) including a single surface EVA.| |\n|Apollo 12| |November 14, 1969, 16:22 GMT Launch Complex 39A|Charles (Pete) Conrad Richard F. Gordon Jr. Alan Bean|Saturn V (SA-507)|\*Yankee Clipper\*|\*Intrepid\*|10 d 04 h 36 m 24 s|First precise Moon landing in Ocean of Storms near Surveyor 3 probe. Two surface EVAs and returned parts of Surveyor to Earth.| |\n|Apollo 13| |April 11, 1970, 19:13 GMT Launch Complex 39A|James Lovell Jack Swigert Fred Haise|Saturn V (SA-508)|\*Odyssey\*|\*Aquarius\*|05 d 22 h 54 m 41 s|Intended Fra Mauro landing cancelled after SM oxygen tank exploded. LM used as \"lifeboat\" for safe crew return. First S-IVB stage impact on Moon for active seismic test.| |\n|Apollo 14| |January 31, 1971, 21:03 GMT Launch Complex 39A|Alan Shepard Stuart Roosa Edgar Mitchell|Saturn V (SA-509)|\*Kitty Hawk\*|\*Antares\*|09 d 00 h 01 m 58 s|Successful Fra Mauro landing. Broadcast first color TV images from lunar surface (other than a few moments at the start of the Apollo 12 moonwalk.) Conducted first materials science experiments in space. Conducted two surface EVAs.| |\n|Apollo 15| |July 26, 1971, 13:34 GMT Launch Complex 39A|David Scott Alfred Worden James Irwin|Saturn V (SA-510)|\*Endeavour\*|\*Falcon\*|12 d 07 h 11 m 53 s|Landing at Hadley-Apennine. First extended LM, three-day lunar stay. First use of Lunar Roving Vehicle. Conducted three lunar surface EVAs and one deep space EVA on return to retrieve orbital camera film from SM.| |\n|Apollo 16| |April 16, 1972, 17:54 GMT Launch Complex 39A|John Young Ken Mattingly Charles Duke|Saturn V (SA-511)|\*Casper\*|\*Orion\*|11 d 01 h 51 m 05 s|Landing in Descartes Highlands. Conducted three lunar EVAs and one deep space EVA.| |\n|Apollo 17| |December 7, 1972, 05:33 GMT Launch Complex 39A|Eugene Cernan Ronald Evans Harrison Schmitt|Saturn V (SA-512)|\*America\*|\*Challenger\*|12d 13 h 51 m 59 s|Landing at Taurus-Littrow. First professional geologist on the Moon. First night launch. Conducted three lunar EVAs and one deep space EVA.| |\n\n### Canceled missions\n\nSeveral planned missions of the Apollo program were canceled for a variety of reasons, including changes in technical direction, the Apollo 1 fire, hardware delays, and budget limitations.\n\n- Before the Apollo 1 fire, two crewed Block I spacecraft missions were planned, but then it was decided that the second one would give no more information about the spacecraft performance not obtained from the first, and could not carry out extra activities such as EVA, and was canceled.\n- The Saturn V's all-up testing strategy and relatively good success rate accomplished the first Moon landing on

the sixth flight, leaving ten available for Moon landings through Apollo 20, but waning public interest in the program led to decreased Congressional funding, forcing NASA to economize. First, Apollo 20 was cut to make a Saturn V available to launch the Skylab space station whole instead of building it on-orbit using multiple Saturn IB launches. Eight months later, Apollo 18 and 19 were also cut to further economize, and because of fears of increased chance of failure with a large number of lunar flights.

**Canceled missions**

Mission	Type	Date	Landing site	CDR	CMP	LMP
Apollo 12	H	November 1969	Ocean of Storms	Pete Conrad	Richard F. Gordon Jr.	Alan Bean
Apollo 12	H	November 14, 1969	Ocean of Storms	Pete Conrad	Richard F. Gordon Jr.	Alan Bean
Apollo 13	H	March 1970	Fra Mauro highlands	Alan Shepard	Stuart Roosa	Edgar Mitchell
Apollo 13	H	April 11, 1970	Failed	Jim Lovell	Jack Swigert	Fred Haise
Apollo 14	H	July 1970	Censorinus crater	Jim Lovell	Ken Mattingly	Fred Haise
Apollo 14	H	January 31, 1971	Fra Mauro highlands	Alan Shepard	Stuart Roosa	Edgar Mitchell
Apollo 15	H	November 1970	Littrow crater	David Scott	Alfred Worden	James Irwin
Apollo 15	J	July 26, 1971	Hadley Rille	David Scott	Alfred Worden	James Irwin
Apollo 16	J	April 1971	Tycho crater	John Young	Jack Swigert	Charles Duke
Apollo 16	J	April 16, 1972	Descartes Highlands	John Young	Ken Mattingly	Charles Duke
Apollo 17	J	September 1971	Marius Hills	Gene Cernan	Ronald Evans	Joe Engle
Apollo 17	J	December 7, 1972	Taurus-Littrow	Gene Cernan	Ronald Evans	Harrison Schmitt
Apollo 18	J	February 1972	Schroter's Valley	Richard F. Gordon Jr.	Vance Brand	Harrison Schmitt
Apollo 19	J	July 1972	Hyginus Rille	Fred Haise	William Pogue	Gerald Carr
Apollo 20	J	December 1972	Copernicus crater	Stuart Roosa	Don L. Lind	Jack Lousma

See also

There were two NASA post-Apollo crewed spaceflight programs that used Apollo hardware:

- Skylab S
- Mission designations – space laboratory missions lasting up to 83 days
- Apollo-Soyuz – first joint US / Soviet crewed spaceflight

Notes

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- This article incorporates public domain material from websites or documents of the National Aeronautics and Space Administration.

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27, 2022, at the Wayback Machine\n- Project Apollo Flickr Photo Archive\n- Interactive Apollo Flag Locations Map\n\nCategories: - Apollo program missions\n- Apollo program\n- NASA missions to the Moon\n- Lists of space missions\n- Crewed missions to the Moon", "title": "List of Apollo missions - Wikipedia", "url": "https://en.wikipedia.org/wiki/List\_of\_Apollo\_missions" }, { "snippet": "\*\*Apollo program\*\*\n| | |\n|--|--|\n|Program overview|Program overview|\n|Country|United States|\n|Organization|NASA|\n|Purpose|Crewed lunar landing|\n|Status|Completed|\n|Program history|Program history|\n|Cost|- \$25.4 billion (1973) - \$257 billion (2020)|\n|Duration|1961-1972|\n|First flight|- SA-1 - October 27, 1961|\n|First crewed flight|- Apollo 7 - October 11, 1968|\n|Last flight|- Apollo 17 - December 19, 1972|\n|Successes|32|\n|Failures|2 (Apollo 1 and 13)|\n|Partial failures|1 (Apollo 6)|\n|Launch sites|- Cape Kennedy - Kennedy Space Center - White Sands|\n|Vehicle information|Vehicle information|\n|Crewed vehicles|- Apollo CSM - Apollo LM|\n|Launch vehicles|- Little Joe II - Saturn I - Saturn IB - Saturn V|\n\nThe \*\*Apollo program\*\*, also known as \*\*Project Apollo\*\*, was the United States human spaceflight program led by NASA, which landed the first humans on the Moon in 1969. Apollo was conceived in 1960 in the Dwight D. Eisenhower presidency during Project Mercury and executed after Project Gemini. Apollo was later dedicated to President John F. Kennedy's national goal, \"before this decade is out, of landing a man on the Moon and returning him safely to the Earth\" in his address to the U.S. Congress on May 25, 1961.\n\nKennedy's goal was accomplished on the Apollo 11 mission, when astronauts Neil Armstrong and Buzz Aldrin landed their Apollo Lunar Module (LM) on July 20, 1969, and walked on the lunar surface, while Michael Collins remained in lunar orbit in the command and service module (CSM), and all three landed safely on Earth in the Pacific Ocean on July 24. Approximately 650 million people worldwide watched this first landing on television. Five subsequent Apollo missions also landed astronauts on the Moon, the last, Apollo 17, in December 1972. In these six spaceflights, twelve people walked on the Moon.\n\nApollo ran from 1961 to 1972, with the first crewed flight in 1968. It encountered a major setback in 1967 when the Apollo 1 cabin fire killed the entire crew during a prelaunch test. After the first Moon landing, sufficient flight hardware remained for nine follow-on landings with a plan for extended lunar geological and astrophysical exploration. Budget cuts forced the cancellation of three of these. Five of the remaining six missions achieved landings; but the Apollo 13 landing had to be aborted after an oxygen tank exploded en route to the Moon, crippling the CSM. The crew barely managed a safe return to Earth by using the Lunar Module as a \"lifeboat\" on the return journey. Apollo used the Saturn family of rockets as launch vehicles, which were also used for an Apollo Applications Program, which consisted of Skylab, a space

station that supported three crewed missions in 1973–1974, and the Apollo–Soyuz Test Project, a joint United States–Soviet Union low Earth orbit mission in 1975.

Apollo set several major human spaceflight milestones. It stands alone in sending humans to the lunar surface. Apollo 8 was the first crewed mission to leave low Earth orbit and to orbit another celestial body, and Apollo 11 was the first crewed mission to land humans on one.

Overall, the Apollo program returned 842 pounds (382 kg) of lunar rocks to Earth, greatly contributing to the understanding of the Moon's composition and geological history. The program laid the foundation for NASA's subsequent human spaceflight capability and funded construction of its Johnson Space Center and Kennedy Space Center. Apollo also spurred advances in many areas of technology incidental to rocketry and human spaceflight, including avionics, telecommunications, and computers.

Following the end of the Apollo program, humans would not leave low Earth orbit until the Artemis II flyby of the Moon in 2026, as part of the Artemis program, established as a successor to Apollo in 2017. Artemis intends to return humans to the Moon's surface no earlier than 2028.

### Name

The program was named after the Greek god Apollo by NASA manager Abe Silverstein, who later said, "I was naming the spacecraft like I'd name my baby."

Silverstein chose the name at home one evening, early in 1960, because he felt "Apollo riding his chariot across the Sun was appropriate to the grand scale of the proposed program".

The context of this was that the program focused at its beginning mainly on developing an advanced crewed spacecraft, the Apollo command and service module, succeeding the Mercury program. A lunar landing became the focus of the program only in 1961. Thereafter Project Gemini instead followed the Mercury program to test and study advanced crewed spaceflight technology.

### Background

#### Origin and spacecraft feasibility studies

The Apollo program was conceived during the Eisenhower administration in early 1960, as a follow-up to Project Mercury. While the Mercury capsule could support only one astronaut on a limited Earth orbital mission, Apollo would carry three. Possible missions included ferrying crews to a space station, circumlunar flights, and eventual crewed lunar landings.

In July 1960, NASA Deputy Administrator Hugh L. Dryden announced the Apollo program to industry representatives at a series of Space Task Group conferences. Preliminary specifications were laid out for a spacecraft with a *mission module* cabin separate from the *command module* (piloting and reentry cabin), and a *propulsion and equipment module*. On August 30, a feasibility study competition was announced, and on October 25, three study contracts were awarded to General Dynamics/Convair, General Electric, and the Glenn L. Martin Company. Meanwhile, NASA performed its own in-house spacecraft design studies led by Maxime Faget, to serve as a gauge to judge and monitor the three industry designs.

#### Political pressure builds

In November 1960, John F. Kennedy was elected president after a campaign that promised American superiority over the Soviet

Union in the fields of space exploration and missile defense. Up to the election of 1960, Kennedy had been speaking out against the "missile gap" that he and many other senators said had developed between the Soviet Union and the United States due to the inaction of President Eisenhower. Beyond military power, Kennedy used aerospace technology as a symbol of national prestige, pledging to make the US not "first but, first and, first if, but first period".

Despite Kennedy's rhetoric, he did not immediately come to a decision on the status of the Apollo program once he became president. He knew little about the technical details of the space program, and was put off by the massive financial commitment required by a crewed Moon landing. When Kennedy's newly appointed NASA Administrator James E. Webb requested a 30 percent budget increase for his agency, Kennedy supported an acceleration of NASA's large booster program but deferred a decision on the broader issue.

On April 12, 1961, Soviet cosmonaut Yuri Gagarin became the first person to fly in space, reinforcing American fears about being left behind in a technological competition with the Soviet Union. At a meeting of the US House Committee on Science and Astronautics one day after Gagarin's flight, many congressmen pledged their support for a crash program aimed at ensuring that America would catch up. Kennedy was circumspect in his response to the news, refusing to make a commitment on America's response to the Soviets.

On April 20, Kennedy sent a memo to Vice President Lyndon B. Johnson, asking Johnson to look into the status of America's space program, and into programs that could offer NASA the opportunity to catch up. Johnson responded approximately one week later, concluding that "we are neither making maximum effort nor achieving results necessary if this country is to reach a position of leadership." His memo concluded that a crewed Moon landing was far enough in the future that it was likely the United States would achieve it first.

On May 25, 1961, twenty days after the first American crewed spaceflight \*Freedom 7\*, Kennedy proposed the crewed Moon landing in a \*Special Message to the Congress on Urgent National Needs\*:

> Now it is time to take longer strides—time for a great new American enterprise—time for this nation to take a clearly leading role in space achievement, which in many ways may hold the key to our future on Earth.

> ... I believe that this nation should commit itself to achieving the goal, before this decade is out, of landing a man on the Moon and returning him safely to the Earth. No single space project in this period will be more impressive to mankind, or more important in the long-range exploration of space; and none will be so difficult or expensive to accomplish.

## NASA expansion

At the time of Kennedy's proposal, only one American had flown in space—less than a month earlier—and NASA had not yet sent an astronaut into orbit. Even some NASA employees doubted whether Kennedy's ambitious goal could be met. By 1963, Kennedy even came close to agreeing to a joint US–USSR Moon mission, to eliminate duplication of effort.

With the clear goal of a crewed landing replacing the more nebulous goals of space

stations and circumlunar flights, NASA decided that, in order to make progress quickly, it would discard the feasibility study designs of Convair, GE, and Martin, and proceed with Faget's command and service module design. The mission module was determined to be useful only as an extra room, and therefore unnecessary. They used Faget's design as the specification for another competition for spacecraft procurement bids in October 1961. On November 28, 1961, North American Aviation won the contract, although its bid was not rated as good as the Martin proposal. Webb, Dryden and Robert Seamans chose it in preference due to North American's longer association with NASA and its predecessor.

Landing humans on the Moon by the end of 1969 required the most sudden burst of technological creativity, and the largest commitment of resources (\$25 billion; \$187 billion in 2024 US dollars) ever made by any nation in peacetime. At its peak, the Apollo program employed 400,000 people and required the support of over 20,000 industrial firms and universities.

On July 1, 1960, NASA established the Marshall Space Flight Center (MSFC) in Huntsville, Alabama. MSFC designed the heavy lift-class Saturn launch vehicles, which would be required for Apollo.

### Manned Spacecraft Center

It became clear that managing the Apollo program would exceed the capabilities of Robert R. Gilruth's Space Task Group, which had been directing the nation's crewed space program from NASA's Langley Research Center. So Gilruth was given authority to grow his organization into a new NASA center, the Manned Spacecraft Center (MSC). A site was chosen in Houston, Texas, on land donated by Rice University, and Administrator Webb announced the conversion on September 19, 1961. It was also clear NASA would soon outgrow its practice of controlling missions from its Cape Canaveral Air Force Station launch facilities in Florida, so a new Mission Control Center would be included in the MSC.

In September 1962, by which time two Project Mercury astronauts had orbited the Earth, Gilruth had moved his organization to rented space in Houston, and construction of the MSC facility was under way, Kennedy visited Rice to reiterate his challenge in a famous speech:

> But why, some say, the Moon? Why choose this as our goal? And they may well ask, why climb the highest mountain? Why, 35 years ago, fly the Atlantic? ...

> We choose to go to the Moon. We choose to go to the Moon in this decade and do the other things, not because they are easy, but because they are hard; because that goal will serve to organize and measure the best of our energies and skills; because that challenge is one that we are willing to accept, one we are unwilling to postpone, and one we intend to win ...

The MSC was completed in September 1963. It was renamed by the United States Congress in honor of Lyndon B. Johnson soon after his death in 1973.

### Launch Operations Center

It also became clear that Apollo would outgrow the Canaveral launch facilities in Florida. The two newest launch complexes were already being built for the Saturn I and IB rockets at the northernmost end: LC-34 and LC-37. An even bigger facility was needed for the

mammoth rocket required for the crewed lunar mission, so land acquisition was started in July 1961 for a Launch Operations Center (LOC) immediately north of Canaveral at Merritt Island.

The design, development and construction of the center was conducted by Kurt H. Debus, a member of Wernher von Braun's original V-2 rocket engineering team. Debus was named the LOC's first Director. Construction began in November 1962. Following Kennedy's death, President Johnson issued an executive order on November 29, 1963, to rename the LOC and Cape Canaveral in honor of Kennedy.

The LOC included Launch Complex 39, a Launch Control Center, and a 130-million-cubic-foot ( $3,700,000 \text{ m}^3$ ) Vertical Assembly Building (VAB). in which the space vehicle (launch vehicle and spacecraft) would be assembled on a mobile launcher platform and then moved by a crawler-transporter to one of several launch pads. Although at least three pads were planned, only two, designated A and B, were completed in October 1965. The LOC also included an Operations and Checkout Building (OCB) to which Gemini and Apollo spacecraft were initially received prior to being mated to their launch vehicles. The Apollo spacecraft could be tested in two vacuum chambers capable of simulating atmospheric pressure at altitudes up to 250,000 feet (76 km), which is nearly a vacuum.

### Organization

Administrator Webb realized that in order to keep Apollo costs under control, he had to develop greater project management skills in his organization, so he recruited George E. Mueller for a high management job. Mueller accepted, on the condition that he have a say in NASA reorganization necessary to effectively administer Apollo. Webb then worked with Associate Administrator (later Deputy Administrator) Seamans to reorganize the Office of Manned Space Flight (OMSF). On July 23, 1963, Webb announced Mueller's appointment as Deputy Associate Administrator for Manned Space Flight, to replace then Associate Administrator D. Brainerd Holmes on his retirement effective September 1. Under Webb's reorganization, the directors of the Manned Spacecraft Center (Gilruth), Marshall Space Flight Center (von Braun), and the Launch Operations Center (Debus) reported to Mueller.

Based on his industry experience on Air Force missile projects, Mueller realized some skilled managers could be found among high-ranking officers in the U.S. Air Force, so he got Webb's permission to recruit General Samuel C. Phillips, who gained a reputation for his effective management of the Minuteman program, as OMSF program controller. Phillips's superior officer Bernard A. Schriever agreed to loan Phillips to NASA, along with a staff of officers under him, on the condition that Phillips be made Apollo Program Director. Mueller agreed, and Phillips managed Apollo from January 1964, until it achieved the first human landing in July 1969, after which he returned to Air Force duty.

Charles Fishman, in *One Giant Leap*, estimated the number of people and organizations involved into the Apollo program as "410,000 men and women at some 20,000 different companies contributed to the effort".

### Choosing a mission mode

Once Kennedy had defined a goal, the Apollo mission planners were faced

with the challenge of designing a spacecraft that could meet it while minimizing risk to human life, limiting cost, and not exceeding limits in possible technology and astronaut skill. Four possible mission modes were considered:

- **Direct Ascent:** The spacecraft would be launched as a unit and travel directly to the lunar surface, without first going into lunar orbit. A 50,000-pound (23,000 kg) Earth return ship would land all three astronauts atop a 113,000-pound (51,000 kg) descent propulsion stage, which would be left on the Moon. This design would have required development of the extremely powerful Saturn C-8 or Nova launch vehicle to carry a 163,000-pound (74,000 kg) payload to the Moon.
- **Earth Orbit Rendezvous (EOR):** Multiple rocket launches (up to 15 in some plans) would carry parts of the Direct Ascent spacecraft and propulsion units for translunar injection (TLI). These would be assembled into a single spacecraft in Earth orbit.
- **Lunar Surface Rendezvous:** Two spacecraft would be launched in succession. The first, an automated vehicle carrying propellant for the return to Earth, would land on the Moon, to be followed some time later by the crewed vehicle. Propellant would have to be transferred from the automated vehicle to the crewed vehicle.
- **Lunar Orbit Rendezvous (LOR):** This turned out to be the winning configuration, which achieved the goal with Apollo 11 on July 20, 1969: a single Saturn V launched a 96,886-pound (43,947 kg) spacecraft that was composed of a 63,608-pound (28,852 kg) Apollo command and service module which remained in orbit around the Moon and a 33,278-pound (15,095 kg) two-stage Apollo Lunar Module spacecraft which was flown by two astronauts to the surface. Its ascent stage was flown back to dock with the command module and was then discarded. Landing the smaller spacecraft on the Moon, and returning an even smaller part (10,042 pounds or 4,555 kilograms) to lunar orbit, minimized the total mass to be launched from Earth, but this was the last method initially considered because of the perceived risk of rendezvous and docking.

In early 1961, direct ascent was generally the mission mode in favor at NASA. Many engineers feared that rendezvous and docking, maneuvers that had not been attempted in Earth orbit, would be nearly impossible in lunar orbit. LOR advocates—including Tom Dolan at Vought and John Houbolt at Langley Research Center—emphasized the important weight reductions that were offered by the LOR approach. Throughout 1960 and 1961, Houbolt campaigned for the recognition of LOR as a viable and practical option. Bypassing the NASA hierarchy, he sent a series of memos and reports on the issue to Associate Administrator Robert Seamans; while acknowledging that he spoke "somewhat as a voice in the wilderness", Houbolt pleaded that LOR should not be discounted in studies of the question.

Seamans's establishment of an ad hoc committee headed by his special technical assistant Nicholas E. Golovin in July 1961, to recommend a launch vehicle to be used in the Apollo program, represented a turning point in NASA's mission mode decision. This committee recognized that the chosen mode was

an important part of the launch vehicle choice, and recommended in favor of a hybrid EOR-LOR mode. Its consideration of LOR—as well as Houbolt's ceaseless work—played an important role in publicizing the workability of the approach.

In late 1961 and early 1962, members of the Manned Spacecraft Center began to come around to support LOR, including the newly hired deputy director of the Office of Manned Space Flight, Joseph Shea, who became a champion of LOR. The engineers at Marshall Space Flight Center (MSFC), who were heavily invested in direct ascent, took longer to become convinced of its merits, but their conversion was announced by Wernher von Braun at a briefing on June 7, 1962.

Even after NASA reached internal agreement, it was far from smooth sailing. Kennedy's science advisor Jerome Wiesner, who had expressed his opposition to human spaceflight to Kennedy before the President took office, and had opposed the decision to land people on the Moon, hired Golovin, who had left NASA, to chair his own "Space Vehicle Panel", ostensibly to monitor, but actually to second-guess NASA's decisions on the Saturn V launch vehicle and LOR by forcing Shea, Seamans, and even Webb to defend themselves, delaying its formal announcement to the press on July 11, 1962, and forcing Webb to still hedge the decision as "tentative".

Wiesner kept up the pressure, even making the disagreement public during a two-day September visit by the President to Marshall Space Flight Center. Wiesner blurted out "No, that's no good" in front of the press, during a presentation by von Braun. Webb jumped in and defended von Braun, until Kennedy ended the squabble by stating that the matter was "still subject to final review". Webb held firm and issued a request for proposal to candidate Lunar Excursion Module (LEM) contractors. Wiesner finally relented, unwilling to settle the dispute once and for all in Kennedy's office, because of the President's involvement with the October Cuban Missile Crisis, and fear of Kennedy's support for Webb. NASA announced the selection of Grumman as the LEM contractor in November 1962.

Space historian James Hansen concludes that:

> Without NASA's adoption of this stubbornly held minority opinion in 1962, the United States may still have reached the Moon, but almost certainly it would not have been accomplished by the end of the 1960s, President Kennedy's target date.

The LOR method had the advantage of allowing the lander spacecraft to be used as a "lifeboat" in the event of a failure of the command ship. Some documents prove this theory was discussed before and after the method was chosen. In 1964 an MSC study concluded, "The LM [as lifeboat] ... was finally dropped, because no single reasonable CSM failure could be identified that would prohibit use of the SPS." That type of failure happened on Apollo 13 when an oxygen tank explosion left the CSM without electrical power. The lunar module provided propulsion, electrical power and life support to get the crew home safely.

## Spacecraft

Faget's preliminary Apollo design employed a cone-shaped command module, supported by one of several service modules providing propulsion and electrical power, sized appropriately

for the space station, cislunar, and lunar landing missions. Once Kennedy's Moon landing goal became official, detailed design began of a command and service module (CSM) in which the crew would spend the entire direct-ascent mission and lift off from the lunar surface for the return trip, after being soft-landed by a larger landing propulsion module. The final choice of lunar orbit rendezvous changed the CSM's role to the translunar ferry used to transport the crew, along with a new spacecraft, the Lunar Excursion Module (LEM, later shortened to LM (Lunar Module) but still pronounced /'lem/) which would take two individuals to the lunar surface and return them to the CSM.

### Command and service module

The command module (CM) was the conical crew cabin, designed to carry three astronauts from launch to lunar orbit and back to an Earth ocean landing. It was the only component of the Apollo spacecraft to survive without major configuration changes as the program evolved from the early Apollo study designs. Its exterior was covered with an ablative heat shield, and had its own reaction control system (RCS) engines to control its attitude and steer its atmospheric entry path. Parachutes were carried to slow its descent to splashdown. The module was 11.42 feet (3.48 m) tall, 12.83 feet (3.91 m) in diameter, and weighed approximately 12,250 pounds (5,560 kg).

A cylindrical service module (SM) supported the command module, with a service propulsion engine and an RCS with propellants, and a fuel cell power generation system with liquid hydrogen and liquid oxygen reactants. A high-gain S-band antenna was used for long-distance communications on the lunar flights. On the extended lunar missions, an orbital scientific instrument package was carried. The service module was discarded just before reentry. The module was 24.6 feet (7.5 m) long and 12.83 feet (3.91 m) in diameter. The initial lunar flight version weighed approximately 51,300 pounds (23,300 kg) fully fueled, while a later version designed to carry a lunar orbit scientific instrument package weighed just over 54,000 pounds (24,000 kg).

North American Aviation won the contract to build the CSM, and also the second stage of the Saturn V launch vehicle for NASA. Because the CSM design was started early before the selection of lunar orbit rendezvous, the service propulsion engine was sized to lift the CSM off the Moon, and thus was oversized to about twice the thrust required for translunar flight. Also, there was no provision for docking with the lunar module. A 1964 program definition study concluded that the initial design should be continued as Block I which would be used for early testing, while Block II, the actual lunar spacecraft, would incorporate the docking equipment and take advantage of the lessons learned in Block I development.

### Apollo Lunar Module

The Apollo Lunar Module (LM) was designed to descend from lunar orbit to land two astronauts on the Moon and take them back to orbit to rendezvous with the command module. Not designed to fly through the Earth's atmosphere or return to Earth, its fuselage was designed totally without aerodynamic considerations and was of an extremely lightweight construction. It consisted of separate descent

and ascent stages, each with its own engine. The descent stage contained storage for the descent propellant, surface stay consumables, and surface exploration equipment. The ascent stage contained the crew cabin, ascent propellant, and a reaction control system. The initial LM model weighed approximately 33,300 pounds (15,100 kg), and allowed surface stays up to around 34 hours. An extended lunar module (ELM) weighed over 36,200 pounds (16,400 kg), and allowed surface stays of more than three days. The contract for design and construction of the lunar module was awarded to Grumman Aircraft Engineering Corporation, and the project was overseen by Thomas J. Kelly.

### Launch vehicles

Before the Apollo program began, Wernher von Braun and his team of rocket engineers had started work on plans for very large launch vehicles, the Saturn series, and the even larger Nova series. In the midst of these plans, von Braun was transferred from the Army to NASA and was made Director of the Marshall Space Flight Center. The initial direct ascent plan to send the three-person Apollo command and service module directly to the lunar surface, on top of a large descent rocket stage, would require a Nova-class launcher, with a lunar payload capability of over 180,000 pounds (82,000 kg). The June 11, 1962, decision to use lunar orbit rendezvous enabled the Saturn V to replace the Nova, and the MSFC proceeded to develop the Saturn rocket family for Apollo.

Since Apollo, like Mercury, used more than one launch vehicle for space missions, NASA used spacecraft-launch vehicle combination series numbers: AS-10x for Saturn I, AS-20x for Saturn IB, and AS-50x for Saturn V (compare Mercury-Redstone 3, Mercury-Atlas 6) to designate and plan all missions, rather than numbering them sequentially as in Project Gemini. This was changed by the time human flights began.

### Little Joe II

Since Apollo, like Mercury, would require a launch escape system (LES) in case of a launch failure, a relatively small rocket was required for qualification flight testing of this system. A rocket bigger than the Little Joe used by Mercury would be required, so the Little Joe II was built by General Dynamics/Convair. After an August 1963 qualification test flight, four LES test flights (A-001 through 004) were made at the White Sands Missile Range between May 1964 and January 1966.

### Saturn I

Saturn I, the first US heavy lift launch vehicle, was initially planned to launch partially equipped CSMs in low Earth orbit tests. The S-I first stage burned RP-1 with liquid oxygen (LOX) oxidizer in eight clustered Rocketdyne H-1 engines, to produce 1,500,000 pounds-force (6,670 kN) of thrust. The S-IV second stage used six liquid hydrogen-fueled Pratt & Whitney RL-10 engines with 90,000 pounds-force (400 kN) of thrust. The S-V third stage flew inactively on Saturn I four times.

The first four Saturn I test flights were launched from LC-34, with only the first stage live, carrying dummy upper stages filled with water. The first flight with a live S-IV was launched from LC-37. This was followed by five launches of boilerplate CSMs (designated AS-101 through AS-105) into orbit in 1964 and 1965. The last three of these further supported the Apollo program by also carrying

Pegasus satellites, which verified the safety of the translunar environment by measuring the frequency and severity of micrometeorite impacts.

In September 1962, NASA planned to launch four crewed CSM flights on the Saturn I from late 1965 through 1966, concurrent with Project Gemini. The 22,500-pound (10,200 kg) payload capacity would have severely limited the systems which could be included, so the decision was made in October 1963 to use the uprated Saturn IB for all crewed Earth orbital flights.

### Saturn IB

The Saturn IB was an upgraded version of the Saturn I. The S-IB first stage increased the thrust to 1,600,000 pounds-force (7,120 kN) by uprating the H-1 engine. The second stage replaced the S-IV with the S-IVB-200, powered by a single J-2 engine burning liquid hydrogen fuel with LOX, to produce 200,000 pounds-force (890 kN) of thrust. A restartable version of the S-IVB was used as the third stage of the Saturn V. The Saturn IB could send over 40,000 pounds (18,100 kg) into low Earth orbit, sufficient for a partially fueled CSM or the LM. Saturn IB launch vehicles and flights were designated with an AS-200 series number, "AS" indicating "Apollo Saturn" and the "2" indicating the second member of the Saturn rocket family.

### Saturn V

Saturn V launch vehicles and flights were designated with an AS-500 series number, "AS" indicating "Apollo Saturn" and the "5" indicating Saturn V. The three-stage Saturn V was designed to send a fully fueled CSM and LM to the Moon. It was 33 feet (10.1 m) in diameter and stood 363 feet (110.6 m) tall with its 96,800-pound (43,900 kg) lunar payload. Its capability grew to 103,600 pounds (47,000 kg) for the later advanced lunar landings. The S-IC first stage burned RP-1/LOX for a rated thrust of 7,500,000 pounds-force (33,400 kN), which was upgraded to 7,610,000 pounds-force (33,900 kN). The second and third stages burned liquid hydrogen; the third stage was a modified version of the S-IVB, with thrust increased to 230,000 pounds-force (1,020 kN) and capability to restart the engine for translunar injection after reaching a parking orbit.

### Astronauts

NASA's director of flight crew operations during the Apollo program was Donald K. "Deke" Slayton, one of the original Mercury Seven astronauts who was medically grounded in September 1962 due to a heart murmur. Slayton was responsible for making all Gemini and Apollo crew assignments.

Thirty-two astronauts were assigned to fly missions in the Apollo program. Twenty-four of these left Earth's orbit and flew around the Moon between December 1968 and December 1972 (three of them twice). Half of the 24 walked on the Moon's surface, though none of them returned to it after landing once. One of the moonwalkers was a trained geologist. Of the 32, Gus Grissom, Ed White, and Roger Chaffee were killed during a ground test in preparation for the Apollo 1 mission.

The Apollo astronauts were chosen from the Project Mercury and Gemini veterans, plus from two later astronaut groups. All missions were commanded by Gemini or Mercury veterans. Crews on all development flights (except the Earth orbit CSM development flights) through the first two landings on Apollo 11 and Apollo 12,

included at least two (sometimes three) Gemini veterans. Harrison Schmitt, a geologist, was the first NASA scientist astronaut to fly in space, and landed on the Moon on the last mission, Apollo 17. Schmitt participated in the lunar geology training of all of the Apollo landing crews.\n\nNASA awarded all 32 of these astronauts its highest honor, the Distinguished Service Medal, given for \"distinguished service, ability, or courage\", and personal \"contribution representing substantial progress to the NASA mission\". The medals were awarded posthumously to Grissom, White, and Chaffee in 1969, then to the crews of all missions from Apollo 8 onward. The crew that flew the first Earth orbital test mission Apollo 7, Walter M. Schirra, Donn Eisele, and Walter Cunningham, were awarded the lesser NASA Exceptional Service Medal, because of discipline problems with the flight director's orders during their flight. In October 2008, the NASA Administrator decided to award them the Distinguished Service Medals. For Schirra and Eisele, this was posthumously.\n\n## Lunar mission

profile\n\nThe first lunar landing mission was planned to proceed:\n\n-  
\*\*Launch\*\* The three Saturn V stages burn for about 11 minutes to achieve a 100-nautical-mile (190 km) circular parking orbit. The third stage burns a small portion of its fuel to achieve orbit.\n-  
\*\*Translunar injection\*\* After one to two orbits to verify readiness of spacecraft systems, the S-IVB third stage reignites for about six minutes to send the spacecraft to the Moon.\n-  
\*\*Transposition and docking\*\* The Spacecraft Lunar Module Adapter (SLA) panels separate to free the CSM and expose the LM. The command module pilot (CMP) moves the CSM out a safe distance, and turns 180°.\n-  
\*\*Extraction\*\* The CMP docks the CSM with the LM, and pulls the complete spacecraft away from the S-IVB. The lunar voyage takes between two and three days. Midcourse corrections are made as necessary using the SM engine.\n-  
\*\*Lunar orbit insertion\*\* The spacecraft passes about 60 nautical miles (110 km) behind the Moon, and the SM engine is fired to slow the spacecraft and put it into a 60-by-170-nautical-mile (110 by 310 km) orbit, which is soon circularized at 60 nautical miles by a second burn.\n-  
After a rest period, the commander (CDR) and lunar module pilot (LMP) move to the LM, power up its systems, and deploy the landing gear. The CSM and LM separate; the CMP visually inspects the LM, then the LM crew move a safe distance away and fire the descent engine for \*\*Descent orbit insertion\*\*, which takes it to a perilune of about 50,000 feet (15 km).\n-  
\*\*Powered descent\*\* At perilune, the descent engine fires again to start the descent. The CDR takes control after pitchover for a vertical landing.\n-  
The CDR and LMP perform one or more EVAs exploring the lunar surface and collecting samples, alternating with rest periods.\n-  
The ascent stage lifts off, using the descent stage as a launching pad.\n-  
The LM rendezvouses and docks with the CSM.\n-  
The CDR and LMP transfer back to the CM with their material samples, then the LM ascent stage is jettisoned, to eventually fall out of orbit and crash on the surface.\n-  
\*\*Trans-Earth injection\*\* The SM engine fires to send the CSM back to Earth.\n-

The SM is jettisoned just before reentry, and the CM turns 180° to face its blunt end forward for reentry. Atmospheric drag slows the CM. Aerodynamic heating surrounds it with an envelope of ionized air which causes a communications blackout for several minutes. Parachutes are deployed, slowing the CM for a splashdown in the Pacific Ocean. The astronauts are recovered and brought to an aircraft carrier.

### Profile variations

The first three lunar missions (Apollo 8, Apollo 10, and Apollo 11) used a free return trajectory, keeping a flight path coplanar with the lunar orbit, which would allow a return to Earth in case the SM engine failed to make lunar orbit insertion. Landing site lighting conditions on later missions dictated a lunar orbital plane change, which required a course change maneuver soon after TLI, and eliminated the free-return option. After Apollo 12 placed the second of several seismometers on the Moon, the jettisoned LM ascent stages on Apollo 12 and later missions were deliberately crashed on the Moon at known locations to induce vibrations in the Moon's structure. The only exceptions to this were the Apollo 13 LM which burned up in the Earth's atmosphere, and Apollo 16, where a loss of attitude control after jettison prevented making a targeted impact. As another active seismic experiment, the S-IVBs on Apollo 13 and subsequent missions were deliberately crashed on the Moon instead of being sent to solar orbit. Starting with Apollo 13, descent orbit insertion was to be performed using the service module engine instead of the LM engine, in order to allow a greater fuel reserve for landing. This was actually done for the first time on Apollo 14, since the Apollo 13 mission was aborted before landing.

## Development history

### Uncrewed flight tests

Two Block I CSMs were launched from LC-34 on suborbital flights in 1966 with the Saturn IB. The first, AS-201 launched on February 26, reached an altitude of 265.7 nautical miles (492.1 km) and splashed down 4,577 nautical miles (8,477 km) downrange in the Atlantic Ocean. The second, AS-202 on August 25, reached 617.1 nautical miles (1,142.9 km) altitude and was recovered 13,900 nautical miles (25,700 km) downrange in the Pacific Ocean. These flights validated the service module engine and the command module heat shield.

A third Saturn IB test, AS-203 launched from pad 37, went into orbit to support design of the S-IVB upper stage restart capability needed for the Saturn V. It carried a nose cone instead of the Apollo spacecraft, and its payload was the unburned liquid hydrogen fuel, the behavior of which engineers measured with temperature and pressure sensors, and a TV camera. This flight occurred on July 5, before AS-202, which was delayed because of problems getting the Apollo spacecraft ready for flight.

### Preparation for crewed flight

Two crewed orbital Block I CSM missions were planned: AS-204 and AS-205. The Block I crew positions were titled Command Pilot, Senior Pilot, and Pilot. The Senior Pilot would assume navigation duties, while the Pilot would function as a systems engineer. The astronauts would wear a modified version of the Gemini spacesuit.

After an uncrewed LM

test flight AS-206, a crew would fly the first Block II CSM and LM in a dual mission known as AS-207/208, or AS-278 (each spacecraft would be launched on a separate Saturn IB). The Block II crew positions were titled Commander, Command Module Pilot, and Lunar Module Pilot. The astronauts would begin wearing a new Apollo A6L spacesuit, designed to accommodate lunar extravehicular activity (EVA). The traditional visor helmet was replaced with a clear "fishbowl" type for greater visibility, and the lunar surface EVA suit would include a water-cooled undergarment.

Deke Slayton, the grounded Mercury astronaut who became director of flight crew operations for the Gemini and Apollo programs, selected the first Apollo crew in January 1966, with Grissom as Command Pilot, White as Senior Pilot, and rookie Donn F. Eisele as Pilot. But Eisele dislocated his shoulder twice aboard the KC135 weightlessness training aircraft, and had to undergo surgery on January 27. Slayton replaced him with Chaffee. NASA announced the final crew selection for AS-204 on March 21, 1966, with the backup crew consisting of Gemini veterans James McDivitt and David Scott, with rookie Russell L. "Rusty" Schweickart. Mercury/Gemini veteran Wally Schirra, Eisele, and rookie Walter Cunningham were announced on September 29 as the prime crew for AS-205.

In December 1966, the AS-205 mission was canceled, since the validation of the CSM would be accomplished on the 14-day first flight, and AS-205 would have been devoted to space experiments and contribute no new engineering knowledge about the spacecraft. Its Saturn IB was allocated to the dual mission, now redesignated AS-205/208 or AS-258, planned for August 1967. McDivitt, Scott and Schweickart were promoted to the prime AS-258 crew, and Schirra, Eisele and Cunningham were reassigned as the Apollo 1 backup crew.

#### Program delays

The spacecraft for the AS-202 and AS-204 missions were delivered by North American Aviation to the Kennedy Space Center with long lists of equipment problems which had to be corrected before flight; these delays caused the launch of AS-202 to slip behind AS-203, and eliminated hopes the first crewed mission might be ready to launch as soon as November 1966, concurrently with the last Gemini mission. Eventually, the planned AS-204 flight date was pushed to February 21, 1967.

North American Aviation was prime contractor not only for the Apollo CSM, but for the Saturn V S-II second stage as well, and delays in this stage pushed the first uncrewed Saturn V flight AS-501 from late 1966 to November 1967. (The initial assembly of AS-501 had to use a dummy spacer spool in place of the stage.)

The problems with North American were severe enough in late 1965 to cause Manned Space Flight Administrator George Mueller to appoint program director Samuel Phillips to head a "tiger team" to investigate North American's problems and identify corrections. Phillips documented his findings in a December 19 letter to NAA president Lee Atwood, with a strongly worded letter by Mueller, and also gave a presentation of the results to Mueller and Deputy Administrator Robert Seamans. Meanwhile, Grumman was also encountering problems with the Lunar Module,

eliminating hopes it would be ready for crewed flight in 1967, not long after the first crewed CSM flights.

### Apollo 1 fire

Grissom, White, and Chaffee decided to name their flight Apollo 1 as a motivational focus on the first crewed flight. They trained and conducted tests of their spacecraft at North American, and in the altitude chamber at the Kennedy Space Center. A "plugs-out" test was planned for January, which would simulate a launch countdown on LC-34 with the spacecraft transferring from pad-supplied to internal power. If successful, this would be followed by a more rigorous countdown simulation test closer to the February 21 launch, with both spacecraft and launch vehicle fueled.

The plugs-out test began on the morning of January 27, 1967, and immediately was plagued with problems. First, the crew noticed a strange odor in their spacesuits which delayed the sealing of the hatch. Then, communications problems frustrated the astronauts and forced a hold in the simulated countdown. During this hold, an electrical fire began in the cabin and spread quickly in the high pressure, 100% oxygen atmosphere. Pressure rose high enough from the fire that the cabin inner wall burst, allowing the fire to erupt onto the pad area and frustrating attempts to rescue the crew. The astronauts were asphyxiated before the hatch could be opened.

NASA immediately convened an accident review board, overseen by both houses of Congress. While the determination of responsibility for the accident was complex, the review board concluded that "deficiencies existed in command module design, workmanship and quality control". At the insistence of NASA Administrator Webb, North American removed Harrison Storms as command module program manager. Webb also reassigned Apollo Spacecraft Program Office (ASPO) Manager Joseph Francis Shea, replacing him with George Low.

To remedy the causes of the fire, changes were made in the Block II spacecraft and operational procedures, the most important of which were use of a nitrogen/oxygen mixture instead of pure oxygen before and during launch, and removal of flammable cabin and space suit materials. The Block II design already called for replacement of the Block I plug-type hatch cover with a quick-release, outward opening door. NASA discontinued the crewed Block I program, using the Block I spacecraft only for uncrewed Saturn V flights. Crew members would also exclusively wear modified, fire-resistant A7L Block II space suits, and would be designated by the Block II titles, regardless of whether a LM was present on the flight or not.

### Uncrewed Saturn V and LM tests

On April 24, 1967, Mueller published an official Apollo mission numbering scheme, using sequential numbers for all flights, crewed or uncrewed. The sequence would start with Apollo 4 to cover the first three uncrewed flights while retiring the Apollo 1 designation to honor the crew, per their widows' wishes.

In September 1967, Mueller approved a sequence of mission types which had to be accomplished in order to achieve the crewed lunar landing. Each step had to be accomplished before the next ones could be performed, and it was unknown how many tries of each mission would be necessary; therefore letters

were used instead of numbers. The **A** missions were uncrewed Saturn V validation; **B** was uncrewed LM validation using the Saturn IB; **C** was crewed CSM Earth orbit validation using the Saturn IB; **D** was the first crewed CSM/LM flight (this replaced AS-258, using a single Saturn V launch); **E** would be a higher Earth orbit CSM/LM flight; **F** would be the first lunar mission, testing the LM in lunar orbit but without landing (a "dress rehearsal"); and **G** would be the first crewed landing. The list of types covered follow-on lunar exploration to include **H** lunar landings, **I** for lunar orbital survey missions, and **J** for extended-stay lunar landings.

The delay in the CSM caused by the fire enabled NASA to catch up on human-rating the LM and Saturn V. Apollo 4 (AS-501) was the first uncrewed flight of the Saturn V, carrying a Block I CSM on November 9, 1967. The capability of the command module's heat shield to survive a trans-lunar reentry was demonstrated by using the service module engine to ram it into the atmosphere at higher than the usual Earth-orbital reentry speed.

Apollo 5 (AS-204) was the first uncrewed test flight of the LM in Earth orbit, launched from pad 37 on January 22, 1968, by the Saturn IB that would have been used for Apollo 1. The LM engines were successfully test-fired and restarted, despite a computer programming error, which cut short the first descent stage firing. The ascent engine was fired in abort mode, known as a "fire-in-the-hole" test, where it was lit simultaneously with jettison of the descent stage. Although Grumman wanted a second uncrewed test, George Low decided the next LM flight would be crewed.

This was followed on April 4, 1968, by Apollo 6 (AS-502) which carried a CSM and a LM Test Article as ballast. The intent of this mission was to achieve trans-lunar injection, followed closely by a simulated direct-return abort, using the service module engine to achieve another high-speed reentry. The Saturn V experienced pogo oscillation, a problem caused by non-steady engine combustion, which damaged fuel lines in the second and third stages. Two S-II engines shut down prematurely, but the remaining engines were able to compensate. The damage to the third stage engine was more severe, preventing it from restarting for trans-lunar injection. Mission controllers were able to use the service module engine to essentially repeat the flight profile of Apollo 4. Based on the good performance of Apollo 6 and identification of satisfactory fixes to the Apollo 6 problems, NASA declared the Saturn V ready to fly crew, canceling a third uncrewed test.

### Crewed development missions

Apollo 7, launched from LC-34 on October 11, 1968, was the C mission, crewed by Schirra, Eisele, and Cunningham. It was an 11-day Earth-orbital flight which tested the CSM systems.

Apollo 8 was planned to be the D mission in December 1968, crewed by McDivitt, Scott and Schweickart, launched on a Saturn V instead of two Saturn IBs. In the summer it had become clear that the LM would not be ready in time. Rather than waste the Saturn V on another simple Earth-orbiting mission, ASPO Manager George Low suggested the

bold step of sending Apollo 8 to orbit the Moon instead, deferring the D mission to the next mission in March 1969, and eliminating the E mission. This would keep the program on track. The Soviet Union had sent two tortoises, mealworms, wine flies, and other lifeforms around the Moon on September 15, 1968, aboard Zond 5, and it was believed they might soon repeat the feat with human cosmonauts. The decision was not announced publicly until completion of Apollo 7. Gemini veterans Frank Borman and Jim Lovell, and rookie William Anders captured the world's attention by making ten lunar orbits in 20 hours, transmitting television pictures of the lunar surface on Christmas Eve, and returning safely to Earth.

The following March, LM flight, rendezvous and docking were demonstrated in Earth orbit on Apollo 9, and Schweickart tested the full lunar EVA suit with its portable life support system (PLSS) outside the LM. The F mission was carried out on Apollo 10 in May 1969 by Gemini veterans Thomas P. Stafford, John Young and Eugene Cernan. Stafford and Cernan took the LM to within 50,000 feet (15 km) of the lunar surface.

The G mission was achieved on Apollo 11 in July 1969 by an all-Gemini veteran crew consisting of Neil Armstrong, Michael Collins and Buzz Aldrin. Armstrong and Aldrin performed the first landing at the Sea of Tranquility at 20:17:40 UTC on July 20, 1969. They spent a total of 21 hours, 36 minutes on the surface, and spent 2 hours, 31 minutes outside the spacecraft, walking on the surface, taking photographs, collecting material samples, and deploying automated scientific instruments, while continuously sending black-and-white television back to Earth. The astronauts returned safely on July 24.

> That's one small step for [a] man, one giant leap for mankind.

– Neil Armstrong, just after stepping onto the Moon's surface

### Production lunar landings

In November 1969, Charles "Pete" Conrad became the third person to step onto the Moon, which he did while speaking more informally than had Armstrong:

> Whoopee! Man, that may have been a small one for Neil, but that's a long one for me.

– Pete Conrad

Conrad and rookie Alan L. Bean made a precision landing of Apollo 12 within walking distance of the Surveyor 3 uncrewed lunar probe, which had landed in April 1967 on the Ocean of Storms. The command module pilot was Gemini veteran Richard F. Gordon Jr. Conrad and Bean carried the first lunar surface color television camera, but it was damaged when accidentally pointed into the Sun. They made two EVAs totaling 7 hours and 45 minutes. On one, they walked to the Surveyor, photographed it, and removed some parts which they returned to Earth.

The contracted batch of 15 Saturn Vs was enough for lunar landing missions through Apollo 20. Shortly after Apollo 11, NASA publicized a preliminary list of eight more planned landing sites after Apollo 12, with plans to increase the mass of the CSM and LM for the last five missions, along with the payload capacity of the Saturn V. These final missions would combine the I and J types in the 1967 list, allowing the CMP to operate a package of lunar orbital sensors and cameras while his companions were on the surface, and

allowing them to stay on the Moon for over three days. These missions would also carry the Lunar Roving Vehicle (LRV) increasing the exploration area and allowing televised liftoff of the LM. Also, the Block II spacesuit was revised for the extended missions to allow greater flexibility and visibility for driving the LRV.

The success of the first two landings allowed the remaining missions to be crewed with a single veteran as commander, with two rookies. Apollo 13 launched Lovell, Jack Swigert, and Fred Haise in April 1970, headed for the Fra Mauro formation. But two days out, a liquid oxygen tank exploded, disabling the service module and forcing the crew to use the LM as a "lifeboat" to return to Earth. Another NASA review board was convened to determine the cause, which turned out to be a combination of damage of the tank in the factory, and a subcontractor not making a tank component according to updated design specifications. Apollo was grounded again, for the remainder of 1970 while the oxygen tank was redesigned and an extra one was added.

#### Mission cutbacks

About the time of the first landing in 1969, it was decided to use an existing Saturn V to launch the Skylab orbital laboratory pre-built on the ground, replacing the original plan to construct it in orbit from several Saturn IB launches; this eliminated Apollo 20. NASA's yearly budget also began to shrink in light of the landing, and NASA also had to make funds available for the development of the upcoming Space Shuttle. By 1971, the decision was made to also cancel missions 18 and 19. The two unused Saturn Vs became museum exhibits at the John F. Kennedy Space Center on Merritt Island, Florida, George C. Marshall Space Center in Huntsville, Alabama, Michoud Assembly Facility in New Orleans, Louisiana, and Lyndon B. Johnson Space Center in Houston, Texas.

The cutbacks forced mission planners to reassess the original planned landing sites in order to achieve the most effective geological sample and data collection from the remaining four missions. Apollo 15 had been planned to be the last of the H series missions, but since there would be only two subsequent missions left, it was changed to the first of three J missions.

Apollo 13's Fra Mauro mission was reassigned to Apollo 14, commanded in February 1971 by Mercury veteran Alan Shepard, with Stuart Roosa and Edgar Mitchell. This time the mission was successful. Shepard and Mitchell spent 33 hours and 31 minutes on the surface, and completed two EVAs totalling 9 hours 24 minutes, which was a record for the longest EVA by a lunar crew at the time.

In August 1971, just after conclusion of the Apollo 15 mission, President Richard Nixon proposed canceling the two remaining lunar landing missions, Apollo 16 and 17. Office of Management and Budget Deputy Director Caspar Weinberger was opposed to this, and persuaded Nixon to keep the remaining missions.

#### Extended missions

Apollo 15 was launched on July 26, 1971, with David Scott, Alfred Worden and James Irwin. Scott and Irwin landed on July 30 near Hadley Rille, and spent just under two days, 19 hours on the surface. In over 18 hours of EVA, they collected about 77 kilograms (170 lb) of lunar material.

Apollo 16

landed in the Descartes Highlands on April 20, 1972. The crew was commanded by John Young, with Ken Mattingly and Charles Duke. Young and Duke spent just under three days on the surface, with a total of over 20 hours EVA.

Apollo 17 was the last of the Apollo program, landing in the Taurus-Littrow region in December 1972. Eugene Cernan commanded Ronald E. Evans and NASA's first scientist-astronaut, geologist Harrison H. Schmitt. Schmitt was originally scheduled for Apollo 18, but the lunar geological community lobbied for his inclusion on the final lunar landing. Cernan and Schmitt stayed on the surface for just over three days and spent just over 23 hours of total EVA.

##### Canceled missions

Several missions were planned for but were canceled before details were finalized.

## Mission summary

Mission	Date	LV	CSM	LM	Crew	Summary	
AS-201	Feb 26, 1966	AS-201	CSM-009	N/a	N/a	First flight of Saturn IB and Block I CSM; suborbital to Atlantic Ocean; qualified heat shield to orbital reentry speed.	
AS-203	Jul 5, 1966	AS-203	N/a	N/a	N/a	No spacecraft; observations of liquid hydrogen fuel behavior in orbit to support design of S-IVB restart capability.	
AS-202	Aug 25, 1966	AS-202	CSM-011	N/a	N/a	Suborbital flight of CSM to Pacific Ocean.	
Apollo 1	Feb 21, 1967	SA-204	CSM-012	N/a	Gus Grissom Ed White Roger B. Chaffee	Not flown. All crew members died in a fire during a launch pad test on January 27, 1967.	
Apollo 4	Nov 9, 1967	SA-501	CSM-017	LTA-10R	N/a	First test flight of Saturn V, placed a CSM in a high Earth orbit; demonstrated S-IVB restart; qualified CM heat shield to lunar reentry speed.	
Apollo 5	Jan 22-23, 1968	SA-204	N/a	LM-1	N/a	Earth orbital flight test of LM, launched on Saturn IB; demonstrated ascent and descent propulsion; human-rated the LM. No crew.	
Apollo 6	Apr 4, 1968	SA-502	CM-020	SM-014	LTA-2R	N/a	Uncrewed, second flight of Saturn V, attempted demonstration of trans-lunar injection, and direct-return abort using SM engine; three engine failures, including failure of S-IVB restart. Flight controllers used SM engine to repeat Apollo 4's flight profile. Human-rated the Saturn V.
Apollo 7	Oct 11-22, 1968	SA-205	CSM-101	N/a	Wally Schirra Walt Cunningham Donn Eisele	First crewed Earth orbital demonstration of Block II CSM, launched on Saturn IB. First live television broadcast from a crewed mission.	
Apollo 8	Dec 21-27, 1968	SA-503	CSM-103	LTA-B	Frank Borman James Lovell William Anders	First crewed flight of Saturn V; First crewed flight to Moon; CSM made 10 lunar orbits in 20 hours.	
Apollo 9	Mar 3-13, 1969	SA-504	CSM-104	*Gumdrop*	LM-3 *Spider*	James McDivitt David Scott Russell Schweickart	Second crewed flight of Saturn V; First crewed flight of CSM and LM in Earth orbit; demonstrated portable life support system to be used on the lunar surface.
Apollo 10	May 18-26, 1969	SA-505	CSM-106	*Charlie Brown*	LM-4 *Snoopy*	Thomas Stafford John Young Eugene Cernan	Dress rehearsal for first lunar landing; flew LM down to 50,000 ft (15 km; 9.5 mi) from lunar surface.
Apollo 11	Jul 16-24, 1969	SA-506	CSM-107	*Columbia*	LM-5 *Eagle*	Neil Armstrong Michael Collins Buzz Aldrin	First landing, in Tranquility

Base, Sea of Tranquility. Surface EVA time: 2h 31m. Samples returned: 47.51 lb (21.55 kg).

|\n|Apollo 12|Nov 14–24, 1969|SA-507|CSM-108 \*Yankee Clipper\*|LM-6 \*Intrepid\*|Pete Conrad Richard Gordon Alan Bean|Second landing, in Ocean of Storms near Surveyor 3. Surface EVA time: 7h 45m. Samples returned: 75.62 lb (34.30 kg).

|\n|Apollo 13|Apr 11–17, 1970|SA-508|CSM-109 \*Odyssey\*|LM-7 \*Aquarius\*|James Lovell Jack Swigert Fred Haise|Third landing attempt aborted in transit to the Moon, due to SM failure. Crew used LM as \"lifeboat\" to return to Earth. Mission called a \"successful failure\".

|\n|Apollo 14|Jan 31 – Feb 9, 1971|SA-509|CSM-110 \*Kitty Hawk\*|LM-8 \*Antares\*|Alan Shepard Stuart Roosa Edgar Mitchell|Third landing, in Fra Mauro formation. Surface EVA time: 9h 21m. Samples returned: 94.35 lb (42.80 kg).

|\n|Apollo 15|Jul 26 – Aug 7, 1971|SA-510|CSM-112 \*Endeavour\*|LM-10 \*Falcon\*|David Scott Alfred Worden James Irwin|Fourth landing, in Hadley–Apennine. First extended mission, used Rover on Moon. Surface EVA time: 18h 33m. Samples returned: 169.10 lb (76.70 kg).

|\n|Apollo 16|Apr 16–27, 1972|SA-511|CSM-113 \*Casper\*|LM-11 \*Orion\*|John Young Ken Mattingly Charles Duke|Fifth landing, in Plain of Descartes. Second extended mission, used Rover on Moon. Surface EVA time: 20h 14m. Samples returned: 207.89 lb (94.30 kg).

|\n|Apollo 17|Dec 7–19, 1972|SA-512|CSM-114 \*America\*|LM-12 \*Challenger\*|Eugene Cernan Ronald Evans Harrison Schmitt|Only Saturn V night launch. Sixth landing, in Taurus–Littrow. Third extended mission, used Rover on Moon. First geologist on the Moon. Apollo's last crewed Moon landing. Surface EVA time: 22h 2m. Samples returned: 243.40 lb (110.40 kg).

|\n\nSource: \*Apollo by the Numbers: A Statistical Reference\* (Orloff 2004).\n\n## Samples returned\n\nThe most famous of the Moon rocks recovered, the Genesis Rock, returned from Apollo 15.\n\nApollo 16's sample 61016, better known as Big Muley, is the largest sample collected during the Apollo program\n\nThe Apollo program returned over 382 kg (842 lb) of lunar rocks and soil to the Lunar Receiving Laboratory in Houston. Today, 75% of the samples are stored at the Lunar Sample Laboratory Facility built in 1979.\n\nThe rocks collected from the Moon are extremely old compared to rocks found on Earth, as measured by radiometric dating techniques. They range in age from about 3.2 billion years for the basaltic samples derived from the lunar maria, to about 4.6 billion years for samples derived from the highlands crust. As such, they represent samples from a very early period in the development of the Solar System, that are largely absent on Earth. One important rock found during the Apollo Program is dubbed the Genesis Rock, retrieved by astronauts David Scott and James Irwin during the Apollo 15 mission. This anorthosite rock is composed almost exclusively of the calcium-rich feldspar mineral anorthite, and is believed to be representative of the highland crust. A geochemical component called KREEP was discovered by Apollo 12, which has no known terrestrial counterpart. KREEP and the anorthositic samples have been used to infer that the outer portion of the Moon was once completely molten (see lunar magma

ocean).

Almost all the rocks show evidence of impact process effects. Many samples appear to be pitted with micrometeoroid impact craters, which is never seen on Earth rocks, due to the thick atmosphere. Many show signs of being subjected to high-pressure shock waves that are generated during impact events. Some of the returned samples are of *impact melt* (materials melted near an impact crater.) All samples returned from the Moon are highly brecciated as a result of being subjected to multiple impact events.

From analyses of the composition of the returned lunar samples, it is now believed that the Moon was created through the impact of a large astronomical body with Earth.

### Costs

Apollo cost \$25.4 billion or approximately \$257 billion (2023) using improved cost analysis.

Of this amount, \$20.2 billion (\$149 billion adjusted) was spent on the design, development, and production of the Saturn family of launch vehicles, the Apollo spacecraft, spacesuits, scientific experiments, and mission operations. The cost of constructing and operating Apollo-related ground facilities, such as the NASA human spaceflight centers and the global tracking and data acquisition network, added an additional \$5.2 billion (\$38.3 billion adjusted).

The amount grows to \$28 billion (\$280 billion adjusted) if the costs for related projects such as Project Gemini and the robotic Ranger, Surveyor, and Lunar Orbiter programs are included.

NASA's official cost breakdown, as reported to Congress in the Spring of 1973, is as follows:

Project Apollo	Cost (original, billion \$)	--	--	Apollo
spacecraft	8.5			
Saturn launch vehicles	9.1			
Launch vehicle engine development	0.9			
Operations	1.7			
<b>Total R&amp;D</b>	<b>20.2</b>			
Tracking and data acquisition	0.9			
Ground facilities	1.8			
Operation of installations	2.5			
<b>Total</b>	<b>25.4</b>			

Accurate estimates of human spaceflight costs were difficult in the early 1960s, as the capability was new and management experience was lacking. Preliminary cost analysis by NASA estimated \$7 billion – \$12 billion for a crewed lunar landing effort. NASA Administrator James Webb increased this estimate to \$20 billion before reporting it to Vice President Johnson in April 1961.

Project Apollo was a massive undertaking, representing the largest research and development project in peacetime. At its peak, it employed over 400,000 employees and contractors around the country and accounted for more than half of NASA's total spending in the 1960s. After the first Moon landing, public and political interest waned, including that of President Nixon, who wanted to rein in federal spending. NASA's budget could not sustain Apollo missions which cost, on average, \$445 million (\$2.73 billion adjusted) each while simultaneously developing the Space Shuttle. The final fiscal year of Apollo funding was 1973.

### Apollo Applications Program

Looking beyond the crewed lunar landings, NASA investigated several post-lunar applications for Apollo hardware. The Apollo Extension Series (*Apollo X*) proposed up to 30 flights to Earth orbit, using the space in the Spacecraft Lunar Module Adapter (SLA) to house a small orbital

laboratory (workshop). Astronauts would continue to use the CSM as a ferry to the station. This study was followed by design of a larger orbital workshop to be built in orbit from an empty S-IVB Saturn upper stage and grew into the Apollo Applications Program (AAP). The workshop was to be supplemented by the Apollo Telescope Mount, which could be attached to the ascent stage of the lunar module via a rack. The most ambitious plan called for using an empty S-IVB as an interplanetary spacecraft for a Venus fly-by mission.

The S-IVB orbital workshop was the only one of these plans to make it off the drawing board. Dubbed Skylab, it was assembled on the ground rather than in space, and launched in 1973 using the two lower stages of a Saturn V. It was equipped with an Apollo Telescope Mount. Skylab's last crew departed the station on February 8, 1974, and the station itself re-entered the atmosphere in 1979 after development of the Space Shuttle was delayed too long to save it.

The Apollo-Soyuz program also used Apollo hardware for the first joint nation spaceflight, paving the way for future cooperation with other nations in the Space Shuttle and International Space Station programs.

### Recent observations

In 2008, Japan Aerospace Exploration Agency's SELENE probe observed evidence of the halo surrounding the Apollo 15 Lunar Module blast crater while orbiting above the lunar surface.

Beginning in 2009, NASA's robotic Lunar Reconnaissance Orbiter, while orbiting 50 kilometers (31 mi) above the Moon, photographed the remnants of the Apollo program left on the lunar surface, and each site where crewed Apollo flights landed. All of U.S. flags left on the Moon during the Apollo missions were found to still be standing, with the exception of the one left during the Apollo 11 mission, which was blown over during that mission's lift-off from the lunar surface; the degree to which these flags retain their original colors remains unknown. The flags cannot be seen through a telescope from Earth.

In a November 16, 2009, editorial, *The New York Times* opined:

[T]here's something terribly wistful about these photographs of the Apollo landing sites. The detail is such that if Neil Armstrong were walking there now, we could make him out, make out his footsteps even, like the astronaut footpath clearly visible in the photos of the Apollo 14 site. Perhaps the wistfulness is caused by the sense of simple grandeur in those Apollo missions. Perhaps, too, it's a reminder of the risk we all felt after the Eagle had landed—the possibility that it might be unable to lift off again and the astronauts would be stranded on the Moon. But it may also be that a photograph like this one is as close as we're able to come to looking directly back into the human past ... There the [Apollo 11] lunar module sits, parked just where it landed 40 years ago, as if it still really were 40 years ago and all the time since merely imaginary.

### Legacy

#### Science and engineering

The Apollo program has been described as the greatest technological achievement in human history. Apollo stimulated many areas of technology, leading to over 1,800 spinoff products as of 2015, including advances in the development of cordless

power tools, fireproof materials, heart monitors, solar panels, digital imaging, and the use of liquid methane as fuel. The flight computer design used in both the lunar and command modules was, along with the Polaris and Minuteman missile systems, the driving force behind early research into integrated circuits (ICs). By 1963, Apollo was using 60 percent of the United States' production of ICs. The crucial difference between the requirements of Apollo and the missile programs was Apollo's much greater need for reliability. While the Navy and Air Force could work around reliability problems by deploying more missiles, the political and financial cost of failure of an Apollo mission was unacceptably high.

Technologies and techniques required for Apollo were developed by Project Gemini. The Apollo project was enabled by NASA's adoption of new advances in semiconductor electronic technology, including metal-oxide-semiconductor field-effect transistors (MOSFETs) in the Interplanetary Monitoring Platform (IMP) and silicon integrated circuit chips in the Apollo Guidance Computer (AGC).

### Cultural impact

The crew of Apollo 8 sent the first live televised pictures of the Earth and the Moon back to Earth, and read from the creation story in the Book of Genesis, on Christmas Eve 1968. An estimated one-quarter of the population of the world saw—either live or delayed—the Christmas Eve transmission during the ninth orbit of the Moon, and an estimated one-fifth of the population of the world watched the live transmission of the Apollo 11 moonwalk.

The Apollo program also affected environmental activism in the 1970s due to photos taken by the astronauts. The most well known include *Earthrise*, taken by William Anders on Apollo 8, and *The Blue Marble*, taken by the Apollo 17 astronauts. *The Blue Marble* was released during a surge in environmentalism, and became a symbol of the environmental movement as a depiction of Earth's frailty, vulnerability, and isolation amid the vast expanse of space.

According to *The Economist*, Apollo succeeded in accomplishing President Kennedy's goal of taking on the Soviet Union in the Space Race by accomplishing a singular and significant achievement, to demonstrate the superiority of the free-market system. The publication noted the irony that in order to achieve the goal, the program required the organization of tremendous public resources within a vast, centralized government bureaucracy.

### Apollo 11 broadcast data restoration project

Prior to Apollo 11's 40th anniversary in 2009, NASA searched for the original videotapes of the mission's live televised moonwalk. After an exhaustive three-year search, it was concluded that the tapes had probably been erased and reused. A new digitally remastered version of the best available broadcast television footage was released instead.

### Depictions on film

### Documentaries

Numerous documentary films cover the Apollo program and the Space Race, including:

- *Footprints on the Moon* (1969)
- *Moonwalk One* (1970)
- *The Greatest Adventure* (1978)
- *For All Mankind* (1989)
- *Moon Shot* (1994 miniseries)
- *"Moon"* from the BBC miniseries *The Planets* (1999)
- *Magnificent Desolation: Walking*

on the Moon 3D\* (2005)\n- \*The Wonder of It All\* (2007)\n- \*In the Shadow of the Moon\* (2007)\n- \*When We Left Earth: The NASA Missions\* (2008 miniseries)\n- \*Moon Machines\* (2008 miniseries)\n- \*James May on the Moon\* (2009)\n- \*NASA's Story\* (2009 miniseries)\n- \*Apollo 11\* (2019)\n- \*Chasing the Moon\* (2019 miniseries)\n\n#### Docudramas\n\nSome missions have been dramatized:\n\n- \*Apollo 13\* (1995)\n- \*Apollo 11\* (1996)\n- \*From the Earth to the Moon\* (1998)\n- \*The Dish\* (2000)\n- \*Space Race\* (2005)\n- \*Moonshot\* (2009)\n- \*First Man\* (2018)\n\n#### Fictional\n\nThe Apollo program has been the focus of several works of fiction, including:\n\n- \*Apollo 18\* (2011), horror movie which was released to negative reviews.\n- \*Transformers: Dark of the Moon\* (2011), Science Fiction/Action movie. The film depicts the Apollo Program as having been created to study and explore a Cybertronian spacecraft known as \"The Ark,\" which crash landed on the dark side of the Moon in the early 1960s.\n- \*Men in Black 3\* (2012), Science Fiction/Comedy movie. Agent J, played by Will Smith, goes back to the Apollo 11 launch in 1969 to ensure that a global protection system is launched in to space.\n- \*For All Mankind\* (2019), TV series depicting an alternate history in which the Soviet Union was the first nation to land a man on the Moon and the Apollo missions were expanded as part of an accelerated Space Race, culminating in the establishment of a permanent US Moon base called Jamestown.\n- \*The Apollo Murders\* (2021), an alternate history novel by Chris Hadfield set in 1973 during the Cold War in which Apollo 18 is launched on a clandestine military mission to the Moon\n- \*Indiana Jones and the Dial of Destiny\* (2023), fifth Indiana Jones film, in which Jürgen Voller, a NASA member and ex-Nazi involved with the Apollo program, wants to time travel. The New York City parade for the Apollo 11 crew is portrayed as a plot point.\n\n## See also\n\n- Apollo 11 in popular culture\n- Apollo program training\n- Apollo Lunar Surface Experiments Package\n- Artemis Program\n- Exploration of the Moon\n- Leslie Cantwell collection\n- List of artificial objects on the Moon\n- List of crewed spacecraft\n- List of missions to the Moon\n- Soviet crewed lunar programs\n- Stolen and missing Moon rocks\n\n## Notes\n\n## References\n\n### Citations\n\n### Sources\n\nThis article incorporates public domain material from websites or documents of the National Aeronautics and Space Administration.\n\n## Further reading\n\n- Gleick, James, \"Moon Fever\" [review of Oliver Morton, \*The Moon: A History of the Future\*; \*Apollo's Muse: The Moon in the Age of Photography\*, an exhibition at the Metropolitan Museum of Art, New York City, July 3 – September 22, 2019; Douglas Brinkley, \*American Moonshot: John F. Kennedy and the Great Space Race\*; Brandon R. Brown, \*The Apollo Chronicles: Engineering America's First Moon Missions\*; Roger D. Launius, \*Reaching for the Moon: A Short History of the Space Race\*; \*Apollo 11\*, a documentary film directed by Todd Douglas Miller; and Michael Collins, \*Carrying the Fire: An Astronaut's Journeys (50th Anniversary Edition)\*], \*The New York Review of Books\*, vol. LXVI, no. 13 (15 August 2019), pp. 54–58.\n\n## External

links\n\nWikimedia Commons has media related to Apollo program.\n\nWikinews has news related to:\n\n\*\*\* Apollo program \*\* \*\n\nLibrary resources about\n\*\*Apollo program\*\*\n\n- Online books\n- Resources in your library\n- Resources in other libraries\n\n- Apollo program history at NASA's Human Space Flight (HSF) website\n- The Apollo Program at the NASA History Program Office\n- The Apollo Program at the National Air and Space Museum\n- Apollo 35th Anniversary Interactive Feature at NASA (in Flash)\n- Lunar Mission Timeline at the Lunar and Planetary Institute\n- Apollo Collection, The University of Alabama in Huntsville Archives and Special Collections\n\n### NASA reports\n- Apollo Program Summary Report (PDF), NASA, JSC-09423, April 1975\n- NASA History Series Publications\n- Project Apollo Drawings and Technical Diagrams at the NASA History Program Office\n- The \*Apollo Lunar Surface Journal\* Archived June 18, 2004, at the Wayback Machine edited by Eric M. Jones and Ken Glover\n- The \*Apollo Flight Journal\* by W. David Woods, et al.\n\n### Multimedia\n- NASA Apollo Program images and videos\n- Apollo Image Archive at Arizona State University\n- Audio recording and transcript of President John F. Kennedy, NASA administrator James Webb, et al., discussing the Apollo agenda (White House Cabinet Room, November 21, 1962)\n- The Project Apollo Archive by Kipp Teague is a large repository of Apollo images, videos, and audio recordings\n- The Project Apollo Archive on Flickr\n- Apollo Image Atlas—almost 25,000 lunar images, Lunar and Planetary Institute\n- Apollo (11, 13 and 17) in real time multimedia project\n- The short film \*The Time of Apollo (1975)\* is available for free viewing and download at the Internet Archive.\n- The short film \*The Time of Apollo (1975)\* is available for free viewing and download at the National Archives.\n- \*The Journeys of Apollo\* – NASA Documentary on YouTube\n- \*Apollo Missions\* (mashup timelapse) on Vimeo\n\nCategories: - Apollo program\n- 1960s in the United States\n- 1970s in the United States\n- Engineering projects\n- American lunar exploration program\n- Human spaceflight programs\n- NASA programs\n- Space program of the United States", "title": "Apollo program - Wikipedia", "url": "https://en.wikipedia.org/wiki/Apollo\_program" }, ], "type": "fetch\_url\_results" }, { "id": "msg\_a5584d41-c74c-4b77-8366-7198e72bc2db", "content": [ { "text": "The Apollo program flew eleven crewed missions (including Apollo 1, which never launched). Each mission progressively tested spacecraft, operations, and ultimately enabled six Moon landings and a dramatic in-flight rescue. [page:1] [page:2]\n\nBelow, “objectives” are what the mission was meant to test or accomplish; “key outcomes” are the main results.\n\n---\n\n## Apollo 1\n\n- \*\*Objectives:\*\* First crewed flight of the Block I Apollo command and service module (CSM) in Earth orbit on a Saturn IB, to validate basic spacecraft systems for later missions. [page:1] \n- \*\*Key outcomes:\*\* Never launched; during a “plugs-out” ground test on January 27, 1967, a cabin fire in a high-pressure

pure oxygen atmosphere killed Gus Grissom, Ed White, and Roger B. Chaffee and destroyed the command module. [page:1] \n - The accident led to major redesigns: mixed nitrogen-oxygen atmosphere on the pad, removal of flammable materials, a quick-opening outward hatch, and cancellation of all crewed Block I flights. [page:1]\n\n---\n\n## Apollo 7\n\n- **Objectives:** First crewed Apollo mission (C-type); shakedown of the redesigned Block II CSM in low Earth orbit using a Saturn IB, including propulsion, life support, navigation, rendezvous capability, and long-duration operations. [page:1] \n- **Key outcomes:** Eleven-day Earth-orbital flight (10 days 20 hours) by Wally Schirra, Donn Eisele, and Walter Cunningham that successfully demonstrated the Block II CSM and made the first live television broadcasts from a U.S. crewed spacecraft. [page:1] \n - Its success gave NASA confidence to commit the next mission, Apollo 8, to the Moon. [web:8][page:1]\n\n---\n\n## Apollo 8\n\n- **Objectives:** C'-type mission; first crewed flight of the Saturn V and first crewed CSM flight to the Moon, to test translunar injection, navigation, communications, and operations in lunar orbit. [page:1] \n- **Key outcomes:** Frank Borman, James Lovell, and William Anders became the first humans to leave Earth orbit, reach the Moon, and orbit it ten times over about 20 hours, then return safely. [web:3][page:1] \n - The crew photographed the lunar far side and the famous "Earthrise," and the mission eliminated the need for a separate medium-Earth-orbit E-type test. [web:3][page:1]\n\n---\n\n## Apollo 9\n\n- **Objectives:** D-type mission; first crewed test of the complete Apollo spacecraft-CSM and Lunar Module (LM)-in low Earth orbit, to validate LM propulsion, rendezvous and docking, and the new lunar EVA suit and Portable Life Support System (PLSS). [web:4][page:1] \n- **Key outcomes:** James McDivitt, David Scott, and Russell "Rusty" Schweickart flew the first crewed LM (\*Spider\*) and CSM (\*Gumdrop\*) together, performed rendezvous and docking maneuvers, and conducted an EVA in the full lunar surface suit using the PLSS. [web:4][web:10][page:1] \n - The mission proved the LM was capable of independent flight and that astronauts could work outside in the suit planned for the Moon. [web:4][page:1]\n\n---\n\n## Apollo 10\n\n- **Objectives:** F-type "dress rehearsal" for the first landing: test all operations of a lunar mission except the actual touchdown-Saturn V performance, lunar orbit operations, LM descent and ascent to near-surface altitude, and high-speed return. [web:5][page:1] \n- **Key outcomes:** Thomas Stafford, John Young, and Eugene Cernan flew CSM \*Charlie Brown\* and LM \*Snoopy\* to the Moon; the LM descended to about 50,000 feet (8.4 nautical miles / ~15.6 km) above the surface before returning to orbit and docking. [web:5][page:1] \n - Demonstrated that the hardware and procedures for a landing were ready, clearing the way for Apollo 11. [web:5][page:1]\n\n---\n\n## Apollo 11\n\n- **Objectives:** G-type mission; first crewed lunar landing to fulfill Kennedy's goal of "landing a man on the Moon and returning him safely to the Earth," testing the full end-to-end profile including descent, landing,

surface EVA, ascent, and sample return. [page:2] \n- **Key outcomes:** On July 20, 1969, Neil Armstrong and Buzz Aldrin landed LM *Eagle* at Tranquility Base in the Sea of Tranquility, while Michael Collins orbited in CSM *Columbia*. [page:1][page:2] \n - Armstrong and Aldrin spent about 21.5 hours on the surface (2.5 hours outside), deployed scientific instruments, and returned 21.55 kg of samples; all three safely splashed down on July 24, 1969. [page:1]

[page:1]\n\n---\n\n## Apollo 12\n\n- **Objectives:** H-type mission; demonstrate a precision landing and expanded surface science with multiple EVAs, including inspecting and retrieving hardware from the earlier Surveyor 3 lander. [page:1] \n- **Key outcomes:** Charles “Pete” Conrad and Alan Bean landed LM *Intrepid* in Ocean of Storms near Surveyor 3, achieving a highly accurate touchdown, while Richard Gordon orbited in CSM *Yankee Clipper*. [page:1] \n - They performed two EVAs (7 h 45 m total), collected 34.30 kg of samples, and returned parts of Surveyor 3 for analysis; the mission also briefly transmitted color TV from the surface before the camera was accidentally pointed at the Sun and failed. [page:1]\n\n---\n\n## Apollo 13\n\n- **Objectives:** H-type mission; planned third landing, targeting the Fra Mauro formation for geological study and further refinement of surface operations. [page:1] \n- **Key outcomes:** An oxygen tank explosion in the service module en route to the Moon crippled CSM *Odyssey*, forcing cancellation of the landing and use of LM *Aquarius* as a “lifeboat” for power, propulsion, and life support. [page:1][page:2] \n - James Lovell, Jack Swigert, and Fred Haise looped around the Moon on a free-return path and returned safely; the mission was later described as a “successful failure” because of the safe recovery and engineering lessons. [page:1]\n\n---\n\n## Apollo 14\n\n- **Objectives:** H-type mission reused Fra Mauro as the landing target (the geology goal from Apollo 13), to study ejecta from the Imbrium impact and conduct improved surface science and long EVAs. [page:1] \n- **Key outcomes:** Alan Shepard and Edgar Mitchell landed LM *Antares* at Fra Mauro while Stuart Roosa orbited in CSM *Kitty Hawk*. [page:1] \n - They spent 33.5 hours on the surface, performed two EVAs totaling about 9 h 21 m, returned 42.80 kg of samples, and were the first to broadcast substantial color TV from the lunar surface. [page:1]\n\n---\n\n## Apollo 15\n\n- **Objectives:** First J-type “extended” mission: longer stay, first Lunar Roving Vehicle (LRV), more sophisticated science payloads in the LM and in the CSM’s Scientific Instrument Module, and intensive geological exploration at the Hadley–Apennine region. [page:1] \n- **Key outcomes:** David Scott and James Irwin landed LM *Falcon* near Hadley Rille while Alfred Worden orbited in CSM *Endeavour*. [page:1] \n - With three EVAs totaling 18 h 33 m, extensive LRV traverses, and 76.70 kg of samples (including the famous “Genesis Rock”), Apollo 15 inaugurated the high-science, long-stay phase of Apollo. [page:1]\n\n---\n\n## Apollo 16\n\n- **Objectives:** Second J-type mission, aimed at the Descartes Highlands to sample what were thought to be ancient highland volcanic

rocks and to extend rover-supported exploration and orbital science. [page:1]

\n- **Key outcomes:** John Young and Charles Duke landed LM *\*Orion\** in the Descartes Highlands while Ken Mattingly remained in CSM *\*Casper\**. [page:1] \n

- They spent nearly three days on the surface with three EVAs totaling 20 h 14 m, drove the LRV on extensive traverses, and returned 94.30 kg of samples, while Mattingly performed a deep-space EVA to retrieve film from the service module. [page:1]

\n\n---\n\n## Apollo 17\n\n- **Objectives:** Final J-type mission and last Apollo lunar landing; target Taurus-Littrow to sample both young volcanic deposits and older highland material, with a professional geologist on the crew for maximum scientific return. [page:1] \n- **Key outcomes:** Eugene Cernan and geologist Harrison Schmitt landed LM *\*Challenger\** in Taurus-Littrow while Ronald Evans piloted CSM *\*America\**; this was the only Saturn V night launch. [page:1]

\n - They spent just over three days on the surface, conducted three EVAs totaling 22 h 2 m with the rover, and returned 110.40 kg of samples; Apollo 17 remains the last crewed mission to the Moon in the Apollo program. [page:1]

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assess if your output fully addresses the query and its subcomponents. Continue until the user query is resolved or until the <tool\_call\_limit> below is reached. End your turn with a comprehensive response. Never mention tool calls in your final response as it would badly impact user experience.

<tool\_call\_limit> Make at most three tool calls before concluding.</tool\_call\_limit></tools\_workflow>

### Citation Instructions

<citation\_instructions> Your response must include at least 1 citation. Add a citation to every sentence that includes information derived from tool outputs. Tool results are provided using `id` in the format `type:index`. `type` is the data source or context. `index` is the unique identifier per citation. <common\_source\_types> are included below.

<common\_source\_types>

- `web`: Internet sources
- `page`: Full web page content
- `conversation\_history`: past queries and answers from your interaction with the user

</common\_source\_types>

<formatting\_citations> Use brackets to indicate citations like this: [type:index]. Commas, dashes, or alternate formats are not valid citation formats. If citing multiple sources, write each citation in a separate bracket like [web:1][web:2][web:3].

Correct: "The Eiffel Tower is in Paris [web:3]."

Incorrect: "The Eiffel Tower is in Paris [web-3]."

</formatting\_citations>

Your citations must be inline – not in a separate References or Citations section. Cite the source immediately after each sentence containing referenced information. If your response presents a markdown table with referenced information from `web`, `memory`, `attached\_file`, or `calendar\_event` tool result, cite appropriately within table cells directly after relevant data instead in of a new column. Do not cite `generated\_image` or `generated\_video` inside table cells.

### Response Guidelines

<response\_guidelines> Responses are displayed on web interfaces where users should not need to scroll extensively. Limit responses to 5 sections maximum. Users can ask follow-up questions if they need additional detail. Prioritize the most relevant information for the initial query.

### Answer Formatting

- Begin with a direct 1-2 sentence answer to the core question.
- Organize the rest of your answer into sections led with Markdown headers (using ##, ###) when appropriate to ensure clarity (e.g. entity definitions, biographies, and wikis).
- Your answer should be at least 3 sentences long.
- Each Markdown header should be concise (less than 6 words) and meaningful.
- Markdown headers should be plain text, not numbered.
- Between each Markdown header is a section consisting of 2-3 well-cited sentences.
- When comparing entities with multiple dimensions, use a markdown table to show differences (instead of lists).
- Whenever possible, present information as bullet point lists to improve readability.
- You are allowed to bold at most one word (**example**) per paragraph. You can't bold consecutive words.
- For grouping multiple related items, present the information with a mix of paragraphs and bullet point lists. Do not nest lists within other

lists.\n\n### Tone\n<tone>\nExplain clearly using plain language. Use active voice and vary sentence structure to sound natural. Ensure smooth transitions between sentences. Avoid personal pronouns like \"I\". Keep explanations direct; use examples or metaphors only when they meaningfully clarify complex concepts that would otherwise be unclear.\n</tone>\n\n### Lists and Paragraphs\n<lists\_and\_paragraphs>\nUse lists for: multiple facts/recommendations, steps, features/benefits, comparisons, or biographical information.\n\nAvoid repeating content in both intro paragraphs and list items. Keep intros minimal. Either start directly with a header and list, or provide 1 sentence of context only.\n\nList formatting:\n- Use numbers when sequence matters; otherwise bullets (-) with a space after the dash.\n- Use numbers when sequence matters; otherwise bullets (-).\n- No whitespace before bullets (i.e. no indenting), one item per line.\n- Sentence capitalization; periods only for complete sentences.\n\nParagraphs:\n- Use for brief context (2-3 sentences max) or simple answers\n- Separate with blank lines\n- If exceeding 3 consecutive sentences, consider restructuring as a list\n</lists\_and\_paragraphs>\n\n### Summaries and Conclusions\n<summaries\_and\_conclusions>\nAvoid summaries and conclusions. They are not needed and are repetitive. Markdown tables are not for summaries. For comparisons, provide a table to compare, but avoid labeling it as 'Comparison/Key Table', provide a more meaningful title.\n</summaries\_and\_conclusions>\n\n## Prohibited Meta-Commentary\n<prohibited\_commentary>\n- Never reference your information gathering process in your final answer.\n- Do not use phrases such as:\n- \"Based on my search results...\"\n- \"Now I have gathered comprehensive information...\"\n- \"According to my research...\"\n- \"My search revealed...\"\n- \"I found information about...\"\n- \"Let me provide a detailed answer...\"\n- \"Let me compile this information...\"\n- \"Short Answer: ...\"\n- Begin answers immediately with factual content that directly addresses the user's query.\n</prohibited\_commentary>\n\n<copyright\_requirements>\n- Never reproduce copyrighted content (text, lyrics, etc.)\n- You may share public domain content (expired copyrights, traditional works)\n- When copyright status is uncertain, treat as copyrighted\n- Keep summaries brief (under 30 words) and original – don't reconstruct sources\n- Brief factual statements (names, dates, facts) are always acceptable\n</copyright\_requirements>\n\nCurrent date: Wednesday, May 27, 2026\n\n", "max\_output\_tokens": 8192, "max\_tool\_calls": null, "metadata": {}, "parallel\_tool\_calls": true, "presence\_penalty": 0, "previous\_response\_id": null, "prompt\_cache\_key": null, "reasoning": null, "safety\_identifier": null, "service\_tier": "default", "store": true, "temperature": 1, "text": { "format": { "type": "text" } }, "tool\_choice": "auto", "tools": [ { "type": "web\_search" }, { "type":

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detail each country's plans to reduce greenhouse gas emissions and contribute to
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Agreement.\nThey detail countries' plans to slash GHG emissions and help limit
global warming to \"well below\" 2 degrees C (3.6 degrees F), with efforts to
limit it to 1.5 degrees C (2.7 degrees F).\nMany NDCs also include measures to
build resilience to climate impacts, such as drought and sea-level rise, and
provide information on the support needed to achieve their commitments.\nUnder
the Paris Agreement, countries agreed to submit new NDCs every five years
reflecting their \"highest possible ambition.\""\nEach round of commitments
should be strengthened based on the latest climate science and countries' own
capabilities and resources.\nMost countries submitted initial emissions targets
prior to adopting the Paris Agreement in 2015.\nThe second round of NDCs, which
set targets through 2030, happened in 2020-2021.\nNow, countries are in the
process of submitting new NDCs with targets that will extend through
2035.\n...\nUnder the Paris Agreement, countries are obligated to have an NDC
and to pursue domestic mitigation measures with the aim of fulfilling their
commitments.\nWhile they are not legally bound to achieve their NDCs, countries
have various responsibilities under the Agreement that are meant to lay the
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new or updated NDC every five years that is more ambitious than its last.",

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Contributions, or NDCs, are countries' self-defined national climate pledges
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international law and global in scope, it not only sets out ambitious global
goals, such as limiting the rise in average global temperature to well below
2 °C compared with pre-industrial levels, but also introduces an innovative
architecture that gives Parties considerable leeway in setting their own climate
change targets.\nIn contrast to common practice under international
environmental law, states' individual contributions are not negotiated at
international level and achievement of set targets is not binding.\n...\nThe
Paris Agreement requires its signatory states (known as Parties) to regularly
formulate their own climate action plans, so-called nationally determined
contributions (NDCs), and to implement measures that help them achieve their
climate action goals.\nThere is, however, no obligation under international law
for Parties to achieve their NDCs.\nParties are, however, required to regularly
report on their progress in implementing their NDCs and the reports are subject
to international peer review.\nIn addition to this Enhanced Transparency
Framework, the Paris Agreement stipulates that Parties must regularly update
their NDCs, that the updated NDCs must not fall short of the targets applicable
prior to the update and that they should reflect the highest possible level of
ambition.\nIn addition, a Global Stocktake is carried out once every five years
to assess the collective progress made towards achieving the long-term
goals.\nThe outcomes of the stocktake are to be taken into account when

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developing nationally determined contributions.", "title": "Global
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goal (“well below” 2°C, with efforts toward 1.5°C), then builds a cyclical
system where countries file and strengthen nationally determined contributions
(NDCs), which are periodically reviewed through a global stocktake to ratchet up
ambition over time.[web:3][web:7] \n\n## Temperature goals (2°C / 1.5°C)\n\n-
The Agreement’s core objective is to hold the increase in global average
temperature to *well below* 2°C above pre-industrial levels and to pursue
efforts to limit the increase to 1.5°C.[web:3][web:4] \n- The 1.5°C reference
is framed as a more desirable aspiration because it substantially reduces the
risks and impacts of climate change compared with 2°C warming.[web:5] \n\n##
Nationally determined contributions (NDCs)\n\n- NDCs are each country’s
self-defined national climate plan, setting out how it will reduce
greenhouse gas emissions and often how it will adapt to climate impacts and what
support it needs.[web:2][web:3][web:6] \n- Under the Paris “ambition cycle,”
every Party must have an NDC, submit a new or updated one at least every five
years, and each successive NDC must represent the country’s “highest possible
ambition” and not fall below the previous one.[web:2][web:4][web:7] \n-
Countries are legally required to prepare, communicate and pursue domestic
measures to implement their NDCs, and to report on progress, but they are
not legally bound to actually achieve the specific numerical targets in
their NDCs.[web:4][web:7] \n\n## Five-year ambition cycle\n\n- The Agreement
works on a recurring five-year cycle: countries submit NDCs, implement policies,
report transparently, and then update NDCs in light of new science,
capabilities, and the outcomes of the global stocktake.[web:3][web:6][web:7]
\n- This iterative design replaces top-down negotiated targets with a bottom-up
system where national pledges are progressively strengthened to collectively
align with the long-term temperature goals.[web:7] \n\n## Global stocktake
mechanism\n\n- The global stocktake (GST), established in Article 14, is a
collective assessment held every five years to evaluate overall progress toward
achieving the purpose and long-term goals of the Paris Agreement, including the
temperature limits.[web:1][web:7] \n- The first stocktake was scheduled for
2023 and then recurs every five years; its outputs are meant to inform and shape
the next round of NDCs and other climate actions, encouraging countries to

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increase ambition over time.[web:1][web:7] \n- The GST assesses mitigation, adaptation, and means of implementation and support (finance, technology, capacity-building), and is linked to the Enhanced Transparency Framework under which Parties regularly report and undergo technical review.[web:1][web:3][web:7] \n\n## How the pieces fit together\n\n- The temperature goals define the overall direction (well below 2°C with efforts toward 1.5°C), while NDCs are the main vehicle through which each country specifies its contribution to that collective objective.[web:3][web:4][web:6] \n- The global stocktake and transparency rules create periodic moments of political and technical scrutiny, intended to push Parties to submit more ambitious NDCs in each cycle so that, over time, global emissions pathways converge toward the agreed temperature limits.[web:1][web:2][web:7]",

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 "instructions": "## Abstract\n<role>\n\nYou are an AI assistant developed by Perplexity AI. Given a user's query, your goal is to generate an expert, useful, factually correct, and contextually relevant response by leveraging available tools and conversation history. First, you will receive the tools you can call iteratively to gather the necessary knowledge for your response. You need to use these tools rather than using internal knowledge. Second, you will receive guidelines to format your response for clear and effective presentation. Third, you will receive guidelines for citation practices to maintain factual accuracy and credibility.\n</role>\n\n## Instructions\n<tools_workflow>\nBegin each turn with tool calls to gather information. You must call at least one tool before answering, even if information exists in your knowledge base. Decompose complex user queries into discrete tool calls for accuracy and parallelization. After each tool call, assess if your output fully addresses the query and its subcomponents. Continue until the user query is resolved or until the <tool_call_limit> below is reached. End your turn with a comprehensive response. Never mention tool calls in your final response as it would badly impact user

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experience.\n\n<tool\_call\_limit> Make at most three tool calls before concluding.</tool\_call\_limit>\n\n</tools\_workflow>\n\n## Citation Instructions\n\n<citation\_instructions>\n\nYour response must include at least 1 citation. Add a citation to every sentence that includes information derived from tool outputs.\n\nTool results are provided using `id` in the format `type:index`. `type` is the data source or context. `index` is the unique identifier per citation.\n\n<common\_source\_types> are included below.\n\n<common\_source\_types>\n\n- `web`: Internet sources\n\n- `page`: Full web page content\n\n- `conversation\_history`: past queries and answers from your interaction with the user\n\n</common\_source\_types>\n\n<formatting\_citations>\n\nUse brackets to indicate citations like this: [type:index]. Commas, dashes, or alternate formats are not valid citation formats. If citing multiple sources, write each citation in a separate bracket like [web:1][web:2][web:3].\n\n\nCorrect: \"The Eiffel Tower is in Paris [web:3].\"\n\nIncorrect: \"The Eiffel Tower is in Paris [web-3].\"\n\n</formatting\_citations>\n\n\nYour citations must be inline – not in a separate References or Citations section. Cite the source immediately after each sentence containing referenced information. If your response presents a markdown table with referenced information from `web`, `memory`, `attached\_file`, or `calendar\_event` tool result, cite appropriately within table cells directly after relevant data instead in of a new column. Do not cite `generated\_image` or `generated\_video` inside table cells.\n\n\n## Response Guidelines\n\n<response\_guidelines>\n\nResponses are displayed on web interfaces where users should not need to scroll extensively. Limit responses to 5 sections maximum. Users can ask follow-up questions if they need additional detail. Prioritize the most relevant information for the initial query.\n\n\n### Answer Formatting\n\n- Begin with a direct 1-2 sentence answer to the core question.\n\n- Organize the rest of your answer into sections led with Markdown headers (using ##, ###) when appropriate to ensure clarity (e.g. entity definitions, biographies, and wikis).\n\n- Your answer should be at least 3 sentences long.\n\n- Each Markdown header should be concise (less than 6 words) and meaningful.\n\n- Markdown headers should be plain text, not numbered.\n\n- Between each Markdown header is a section consisting of 2-3 well-cited sentences.\n\n- When comparing entities with multiple dimensions, use a markdown table to show differences (instead of lists).\n\n- Whenever possible, present information as bullet point lists to improve readability.\n\n- You are allowed to bold at most one word (\*\*example\*\*) per paragraph. You can't bold consecutive words.\n\n- For grouping multiple related items, present the information with a mix of paragraphs and bullet point lists. Do not nest lists within other lists.\n\n\n### Tone\n\n<tone>\n\nExplain clearly using plain language. Use active voice and vary sentence structure to sound natural. Ensure smooth transitions between sentences. Avoid personal pronouns like \"I\". Keep explanations direct; use examples or metaphors only when they meaningfully clarify complex concepts

that would otherwise be unclear.

### Lists and Paragraphs

Use lists for: multiple facts/recommendations, steps, features/benefits, comparisons, or biographical information.

Avoid repeating content in both intro paragraphs and list items. Keep intros minimal. Either start directly with a header and list, or provide 1 sentence of context only.

List formatting:

- Use numbers when sequence matters; otherwise bullets (-) with a space after the dash.
- Use numbers when sequence matters; otherwise bullets (-).
- No whitespace before bullets (i.e. no indenting), one item per line.
- Sentence capitalization; periods only for complete sentences.

Paragraphs:

- Use for brief context (2-3 sentences max) or simple answers
- Separate with blank lines
- If exceeding 3 consecutive sentences, consider restructuring as a list

### Summaries and Conclusions

Avoid summaries and conclusions. They are not needed and are repetitive. Markdown tables are not for summaries. For comparisons, provide a table to compare, but avoid labeling it as 'Comparison/Key Table', provide a more meaningful title.

### Prohibited Meta-Commentary

Never reference your information gathering process in your final answer.

Do not use phrases such as:

- "Based on my search results..."
- "Now I have gathered comprehensive information..."
- "According to my research..."
- "My search revealed..."
- "I found information about..."
- "Let me provide a detailed answer..."
- "Let me compile this information..."
- "Short Answer: ..."

Begin answers immediately with factual content that directly addresses the user's query.

### Copyright Requirements

Never reproduce copyrighted content (text, lyrics, etc.)

- You may share public domain content (expired copyrights, traditional works)
- When copyright status is uncertain, treat as copyrighted
- Keep summaries brief (under 30 words) and original – don't reconstruct sources
- Brief factual statements (names, dates, facts) are always acceptable

Current date: Thursday, May 21, 2026

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See [Filters](#) for the full list of available parameters.

To run without tools, set `tools_disabled: true` on the request. Passing `tools: []` does **not** clear preset tools. An empty array is treated the same as omitting the field, and the preset's defaults still apply.

## Next Steps

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Shape responses with `response_format` and learn the full response payload structure.

Constrain search with domain, recency, and region parameters.

Configure the `web_search` tool for source-grounded context.

Choose a preset that matches your latency, depth, and tool requirements.